CS1231S Discrete Structures

WEEK 11 RECITATION
LECTURE 12. GRAPHS

29 – 30 October 2026

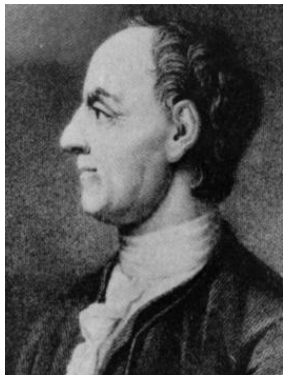
Aaron Tan



CS1231S (AY2026/27 Sem1) QnA



"The origins of graph theory are humble, even frivolous." ~*Norman L. Biggs*



The Father of Graph Theory
Leonhard Euler (1707-1783)



Credit: Wikipedia

The 7 bridges of Königsberg

Euler's formula: F + V = E + 2

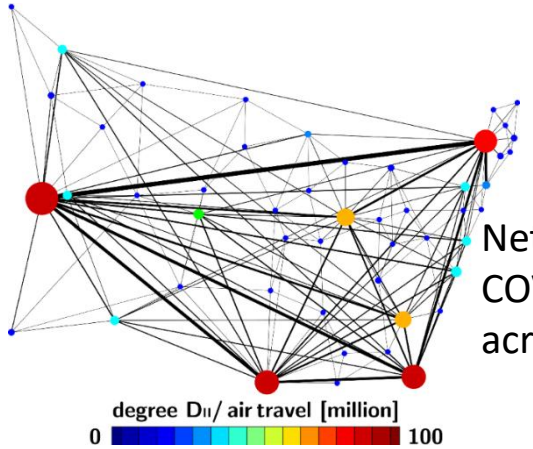
Knight's Tour Problem

Travelling Salesman Problem

Applications:

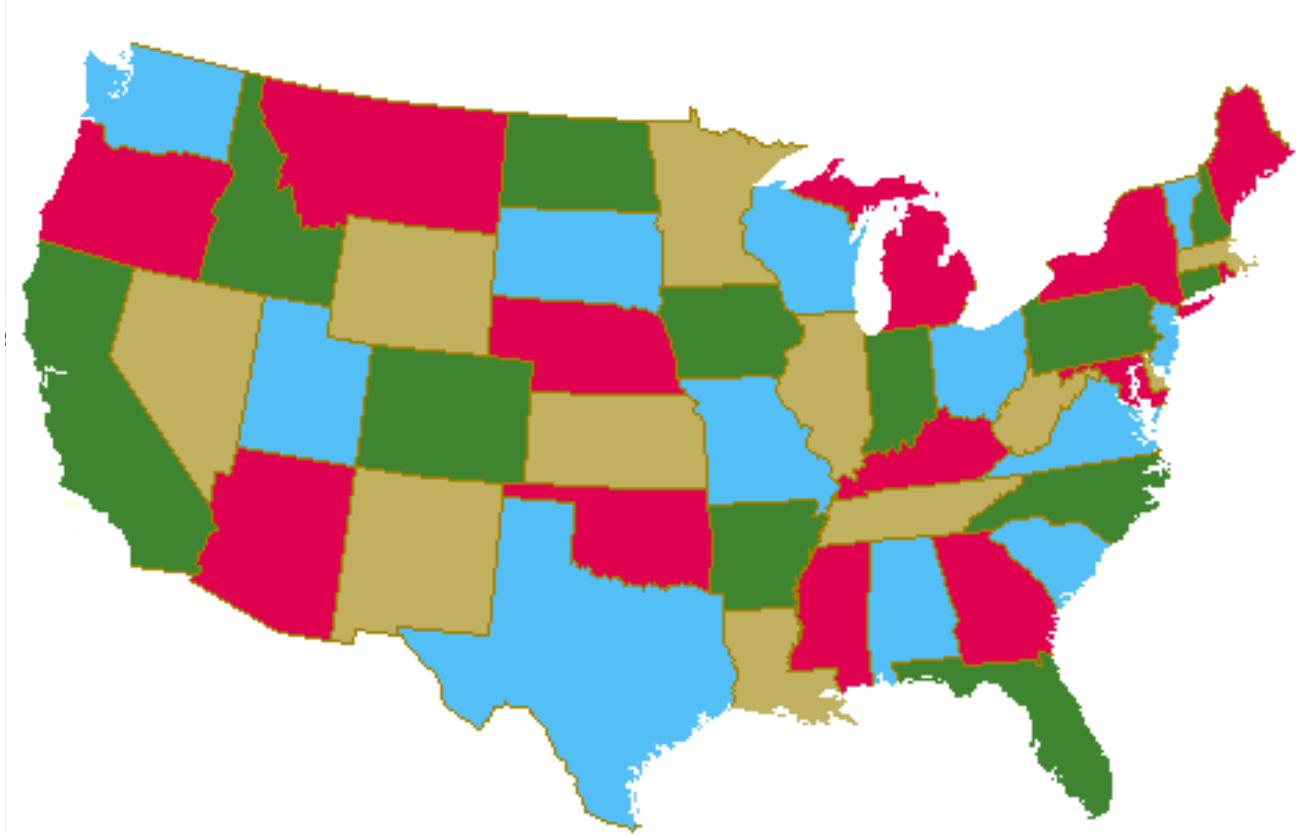
- CS: hardware, data structures, image processing, data mining, network design, etc.
- GPS to find shortest path
- Ranking hyperlinks in search engines
- Social network analysis
- DNA sequence
- …

Graph as an excellent modelling tool…



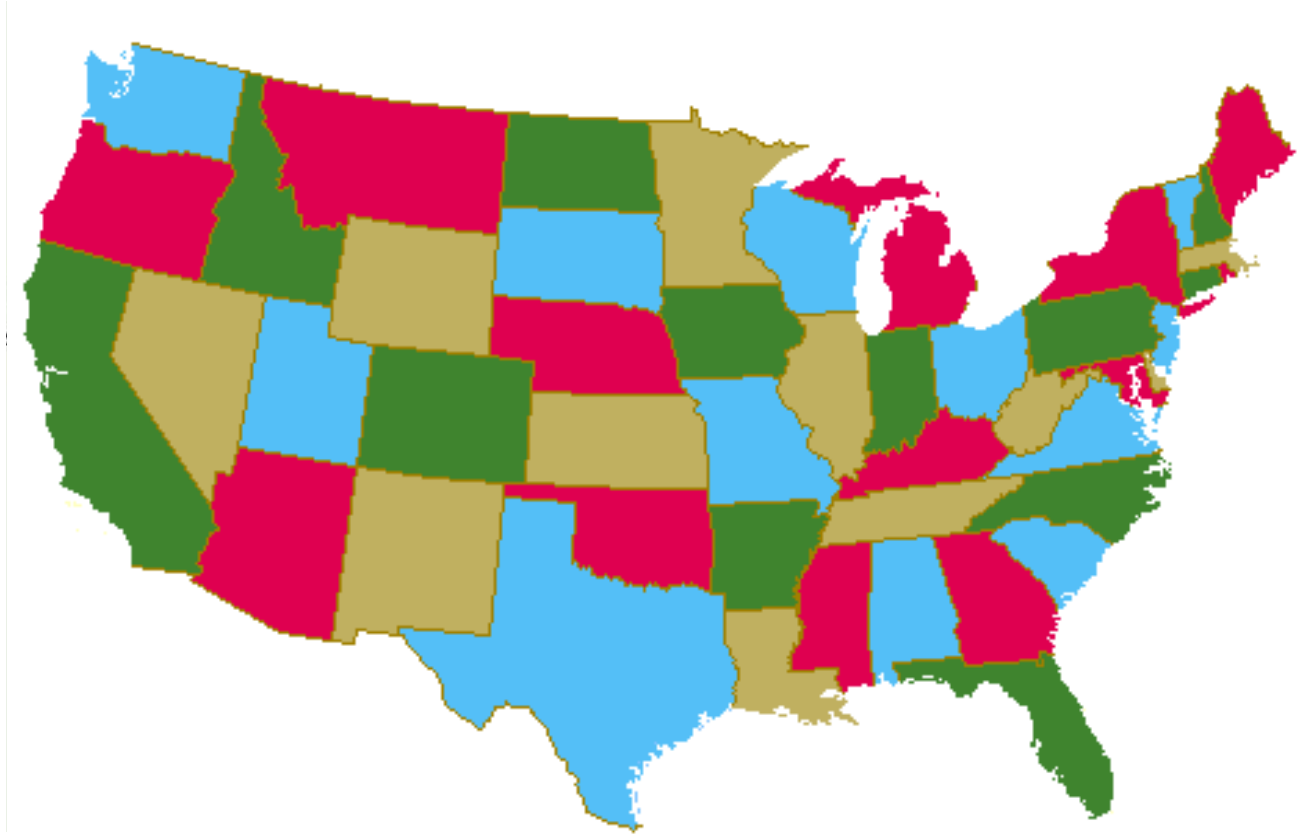
Network model of COVID-19 spreading across the United States

Shall we add some colours to this map of the United States?



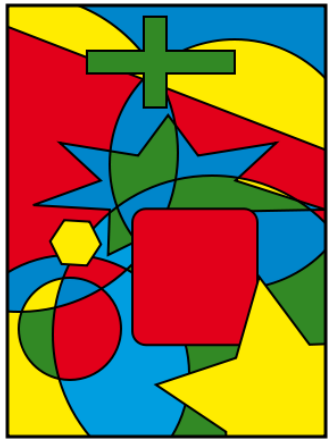
- Four-Colour Conjecture
  - Proposed by Guthrie in 1852, who conjectured that…
  - Four colours are sufficient to colour any map in a plane, such that regions that share a common boundary do not share the same colour.
  - Many false proofs since then.
  - Finally proved by Appel and Haken in 1977, with the help of computer.
  - Robertson et al. provided another proof in 1996.

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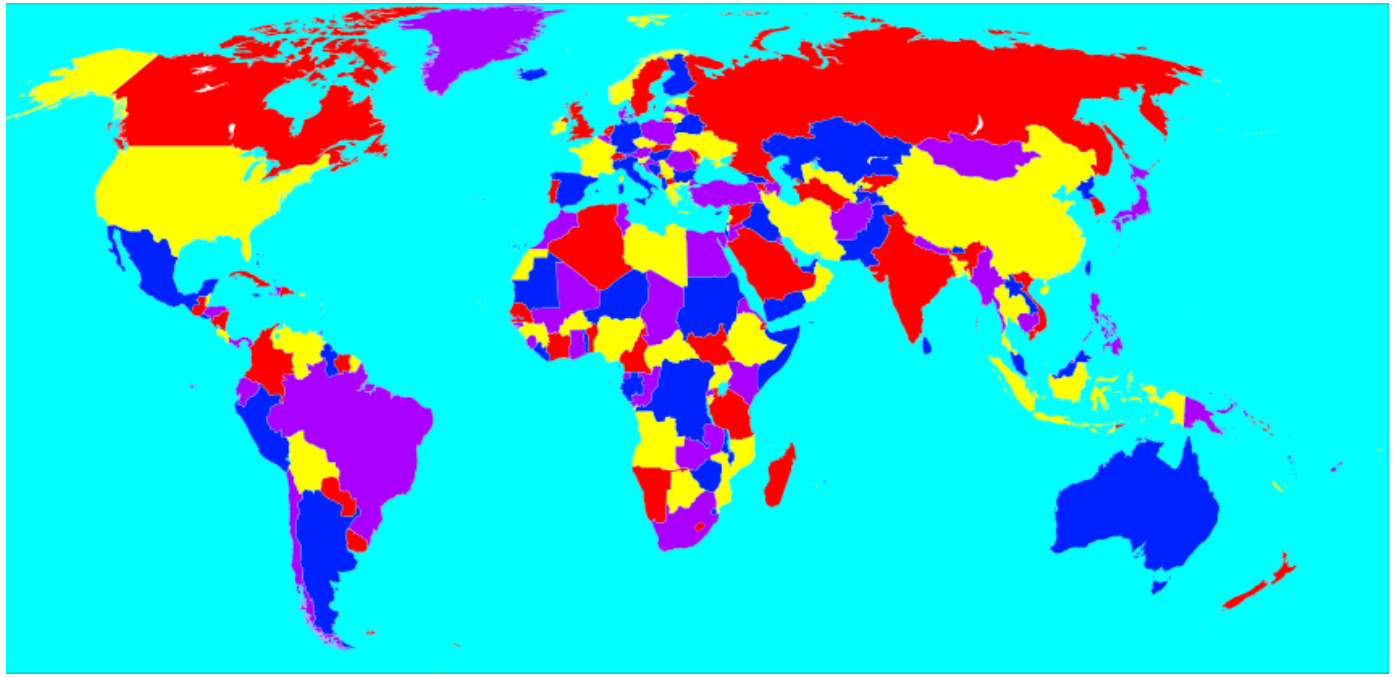


- Four-Colour Conjecture
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- But this is a map, not a graph!
- However, we can model it as a graph.
- But what is a graph?



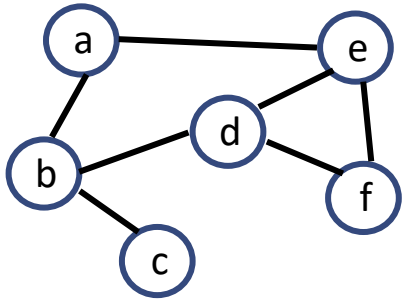
Example of a 4-coloured map.



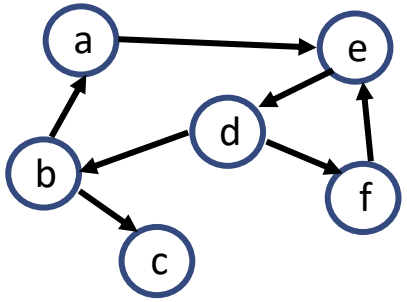
World map with 4 colours.

Types of graphs (informal intro):

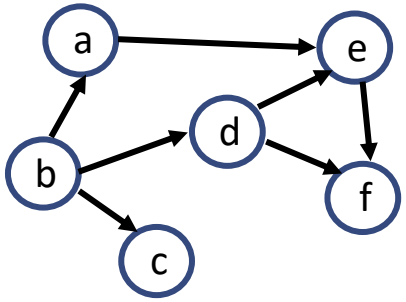
Graphs are mathematical structures used to model pairwise relations between objects.



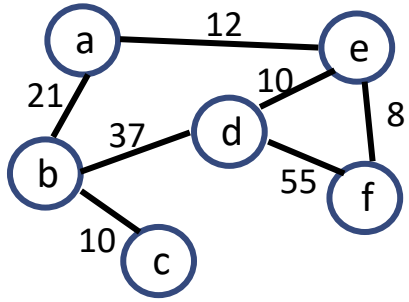
Undirected graph



Directed graph (Digraph)



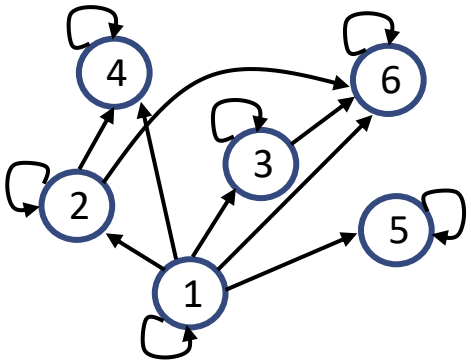
Directed acyclic graph (DAG)



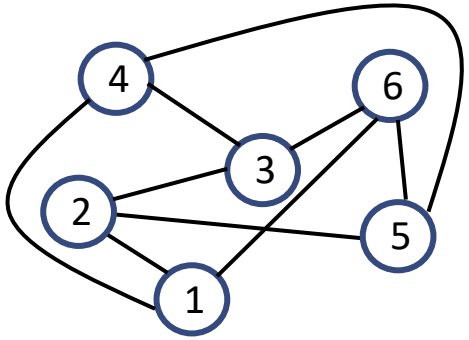
Weighted graph

Bet you have seen graphs earlier in CS1231S before!

Relation R on a set $A = \{1,2,3,4,5,6\}$, s.t. $\forall x, y \in A$, $x \ R \ y$ iff $x \mid y$.



Relation T on a set $A = \{1,2,3,4,5,6\}$, s.t. $\forall x, y \in A$, $x \ T \ y$ iff $(x + y) = 2k + 1$ for some $k \in \mathbb{Z}$.

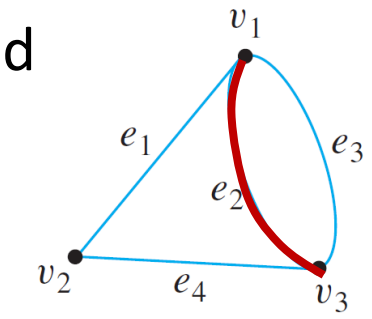


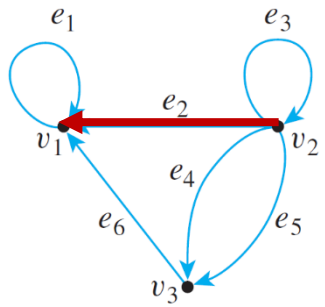
- An undirected graph is denoted by $G = (V, E)$ where
  - $V = \{v_1, v_2, \cdots, v_n\}$ is the set of vertices (or nodes) in G; and
  - $E = \{e_1, e_2, \cdots, e_k\}$ is the set of undirected edges in G.
  - An undirected edge e connecting v_i and v_j is denoted as $e = \{v_i, v_j\}$.

- A directed graph is denoted by $G = (V, E)$ where
  - $V = \{v_1, v_2, \cdots, v_n\}$ is the set of vertices (or nodes) in G; and
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  - A directed edge e connecting v_i and v_j is denoted as $e = (v_i, v_j)$.

Undirected graph

$e_2 = \{v_1, v_3\}$





Directed graph

$e_2 = (v_2, v_1)$

Modelling Graph Problems



Map Colouring Problem

Solve it as a graph problem.

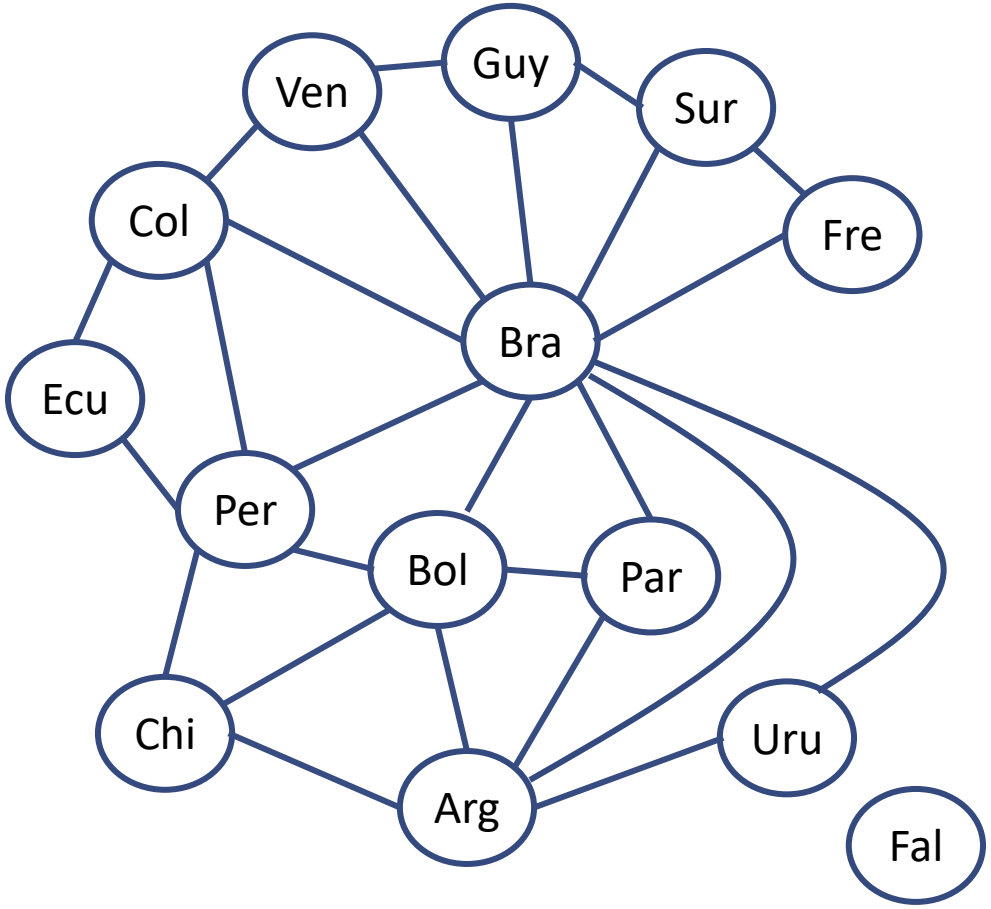
Draw a graph in which the vertices represent the states, with every edge joining two vertices represents the states sharing a common border.

Such two vertices cannot be coloured with the same colour.

A **vertex colouring** of a graph is an assignment of colours to vertices so that no two adjacent vertices have the same colour.

Modelling Graph Problems

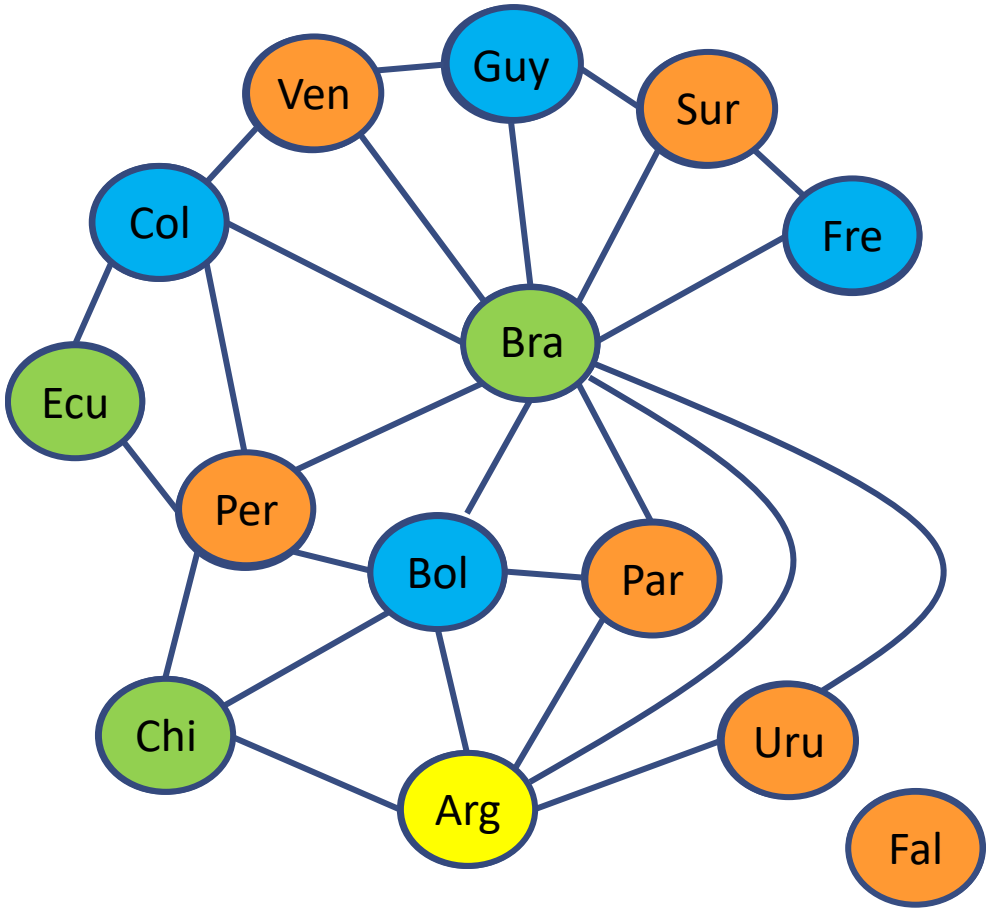




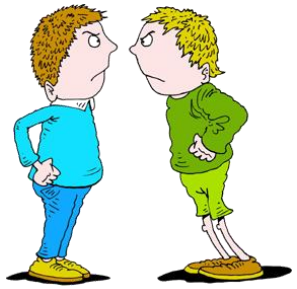
Modelling Graph Problems



4 colours used!



Modelling Graph Problems



Wedding Planner

You are your best friend's wedding planner and you need to plan the seating arrangement for his 16 guests attending his wedding dinner. However, some of the guests cannot get along with some others.

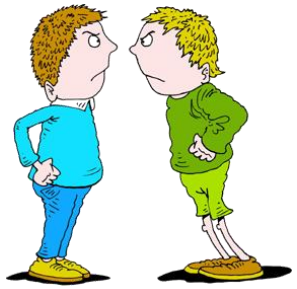
- *A* doesn't get along with *F*, *G* or *H*.
- *B* doesn't get along with *C*, *D* or *H*.
- *C* doesn't get along with *B*, *D*, *E*, *G* or *H*.
- *D* doesn't get along with *B*, *C* or *E*.
- *E* doesn't get along with *C*, *D*, *F*, or *G*.
- *F* doesn't get along with *A*, *E* or *G*.
- *G* doesn't get along with *A*, *C*, *E* or *F*.
- *H* doesn't get along with *A*, *B* or *C*.

You don't want to put guests who cannot get along with each other at the same table!

How many tables do you need?

Acknowledgement: http://www.math.uri.edu/~eaton/0131873814_MEb.pdf

Modelling Graph Problems



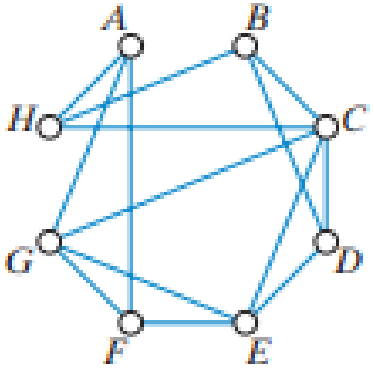
Wedding Planner

Graph with vertices representing the guests, and an edge is drawn between two guests who don't get along.

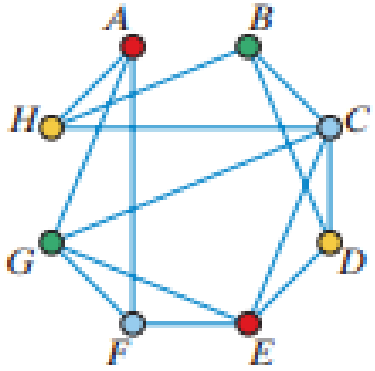
- *A* doesn't get along with *F*, *G* or *H*.
- *B* doesn't get along with *C*, *D* or *H*.
- *C* doesn't get along with *B*, *D*, *E*, *G* or *H*.
- *D* doesn't get along with *B*, *C* or *E*.
- *E* doesn't get along with *C*, *D*, *F*, or *G*.
- *F* doesn't get along with *A*, *E* or *G*.
- *G* doesn't get along with *A*, *C*, *E* or *F*.
- *H* doesn't get along with *A*, *B* or *C*.

Vertex colouring problem.
4 colours (4 tables)?

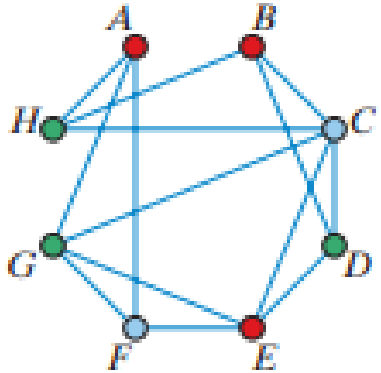
3 colours
(3 tables)!



(a)



(b)



(c)

Acknowledgement: http://www.math.uri.edu/~eaton/0131873814_MEb.pdf

Modelling Graph Problems

Other vertex colouring problems

	If the vertices represent…	And two vertices are adjacent if ….	Then a **vertex colouring** can be used to…
1.	classes,	the corresponding classes have students in common,	schedule classes.
2.	radio stations,	the stations are close enough to interfere with each other,	assign non-interfering frequencies to the stations.
3.	traffic signals at an intersection,	the corresponding signals cannot be green at the same time,	designate sets of signals that can be green at the same time.

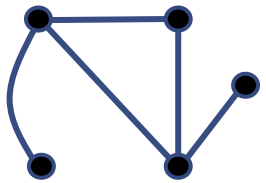
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Simple Graphs

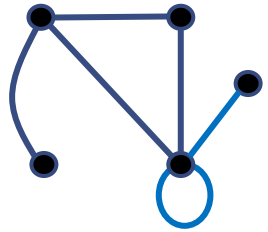
Definition: Simple Graph

A **simple graph** is an undirected graph that does not have any loops or parallel edges. (That is, there is at most one edge between each pair of distinct vertices.)

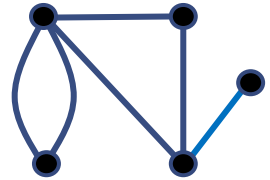
Simple graph



Non simple graph



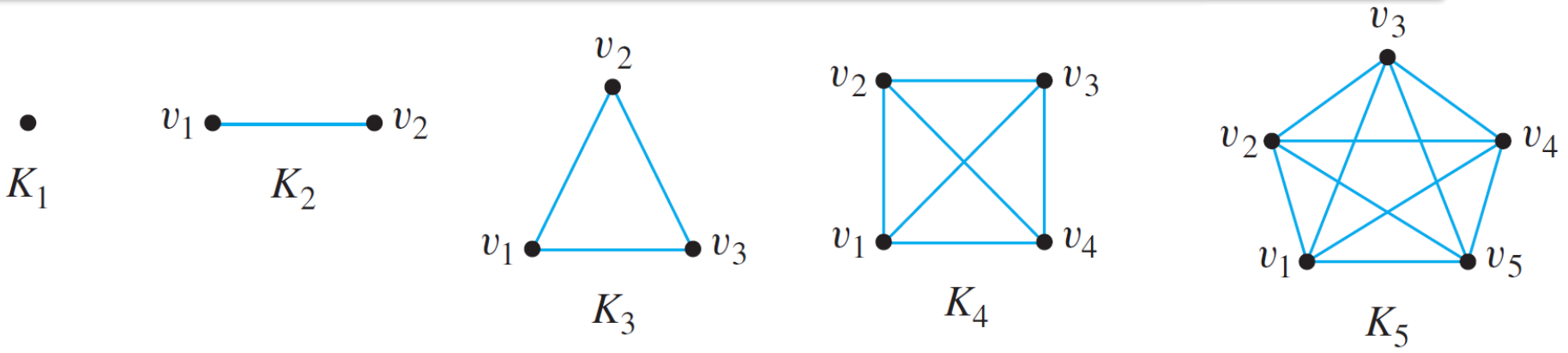
Non simple graph



Complete Graphs

Definition: Complete Graph

A **complete graph** on ***n* vertices**, $n > 0$, denoted $\boldsymbol{K_n}$, is a simple graph with n vertices and exactly one edge connecting each pair of distinct vertices.

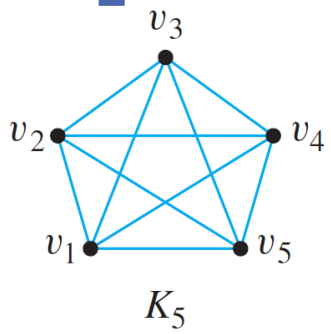


Fun: The notation K_n is rumoured to be used to honour the contributions of Kazimierz Kuratowski to graph theory.

Important fact:

How many edges are there in K_n?

Complete Graphs: Number of Edges



K_5

Claim: Number of edges in a complete graph K_n is $\frac{n(n-1)}{2}$

There exists exactly one edge between 2 distinct vertices:

$$\binom{n}{2} = \frac{n!}{(n-2)!2!} = \frac{n(n-1)}{2}$$

Define $V(G_n)$ to be the number of vertices in a graph G with n vertices.

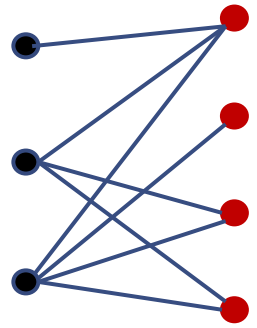
Prove by induction that $V(G_n) = \frac{n(n-1)}{2}$ for all $n \geq 1$.

1. Let $P(n) \equiv V(G_n) = \frac{n(n-1)}{2}$ for all $n \geq 1$.
2. (Basis step) $V(G_1) = 0 = \frac{1(1-1)}{2}$. Hence $P(1)$ is true.
3. (Inductive step) Assume $P(k)$, i.e. for arbitrary $k \geq 1, V(G_k) = \frac{k(k-1)}{2}$
   - 3.1. For a graph with $k+1$ vertices, the $(k+1)^{\text{th}}$ vertex must be adjacent to each of the k vertices.
   - 3.2. Hence, $V(G_{k+1}) = V(G_k) + k = \frac{k(k-1)}{2} + k = \frac{(k+1)k}{2}$
   - 3.3. Hence $P(k+1)$ is true.
4. Therefore, $V(G_n) = \frac{n(n-1)}{2}$ for all $n \geq 1$ by mathematical induction.

Bipartite Graphs and Complete Bipartite Graphs

Definition: Bipartite Graph

A **bipartite graph** is a simple graph whose vertices can be divided into two disjoint sets U and V such that every edge connects a vertex in U to one in V.

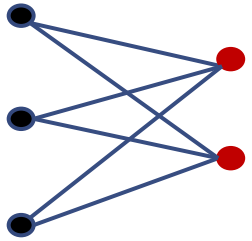


Bipartite graph

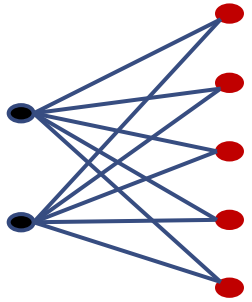
Definition: Complete Bipartite Graph

A **complete bipartite graph** is a bipartite graph on two disjoint sets U and V such that every vertex in U connects to every vertex in V.

If $|U| = m$ and $|V| = n$, the complete bipartite graph is denoted as $K_{m,n}$.



$K_{3,2}$



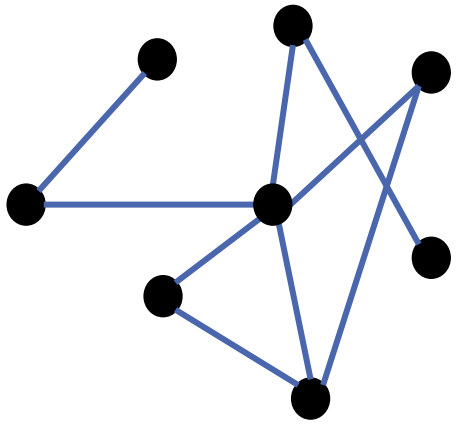
$K_{2,5}$

Subgraph

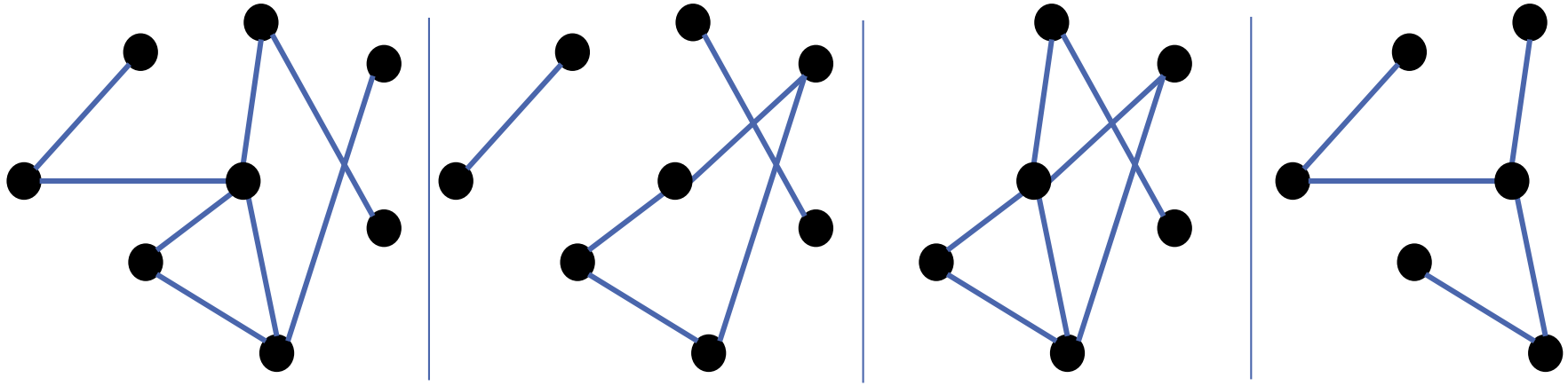
Definition: Subgraph of a Graph

A graph *H* is said to be a **subgraph** of graph *G* if and only if every vertex in *H* is also a vertex in *G*, every edge in *H* is also an edge in *G*, and every edge in *H* has the same endpoints as it has in *G*.

A graph *G*



Some subgraphs of *G*



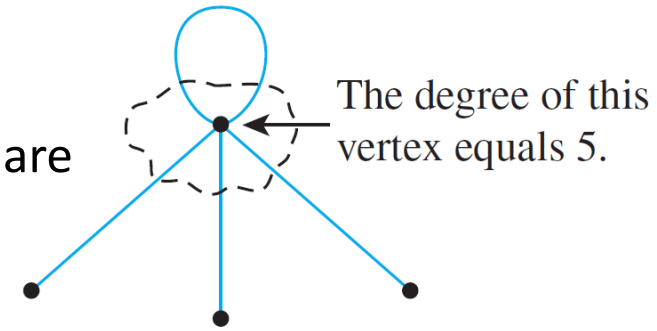
Degree of a Vertex, Undirected Graph

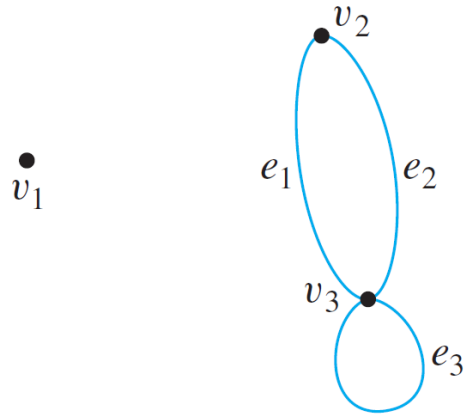
Definition: Degree of a Vertex and Total Degree of an Undirected Graph

Let G be a undirected graph and v a vertex of G. The **degree** of v, denoted **deg(v)**, equals the number of edges that are incident on v, with an edge that is a loop counted twice.

The **total degree of G** is the sum of the degrees of all the vertices of G.

The degree of a vertex can be obtained from the drawing of a graph by counting how many end segments of edges are incident on the vertex.





$\deg(v_1) = 0$
$\deg(v_2) = 2$
$\deg(v_3) = 4$
Total degree = 6

The Handshake Theorem

Theorem 10.1.1 The Handshake Theorem



If G is any graph, then the sum of the degrees of all the vertices of G equals twice the number of edges of G. Specifically, if the vertices of G are $v_1, v_2, \ldots, v_n$, where $n \geq 0$, then

$$\text{The total degree of } G = \deg(v_1) + \deg(v_2) + \ldots + \deg(v_n)$$
$$= 2 \times (\text{the number of edges of } G).$$

Corollary 10.1.2

The total degree of a graph is even.

Proposition 10.1.3

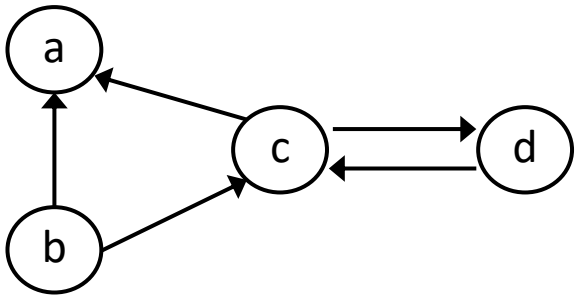
In any graph there are an even number of vertices of odd degree.

Indegree and Outdegree of a Vertex, Directed Graph

Definition: Indegree and Outdegree of a Vertex of a Directed Graph

Let $G=(V,E)$ be a directed graph and v a vertex of G. The **indegree** of v, denoted $\deg^-(v)$, is the number of directed edges that end at v. The **outdegree** of v, denoted $\deg^+(v)$, is the number of directed edges that originate from v.

Note that $$\sum_{v \in V} deg^-(v) = \sum_{v \in V} deg^+(v) = |E|$$

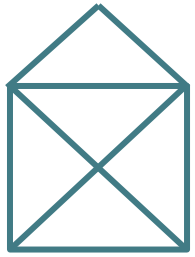


$\deg^-(a) = 2$; $\deg^+(a) = 0$;
$\deg^-(b) = 0$; $\deg^+(b) = 2$;
$\deg^-(c) = 2$; $\deg^+(c) = 2$;
$\deg^-(d) = 1$; $\deg^+(d) = 1$.

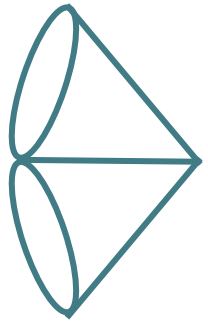
Unicursal Drawing



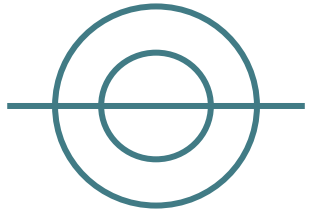
Can you draw the following figures in a single stroke without lifting your pencil or retracing?



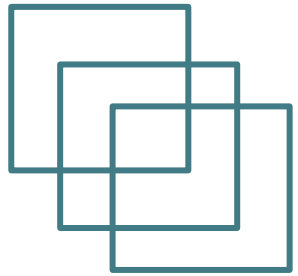
(1)



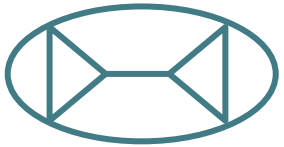
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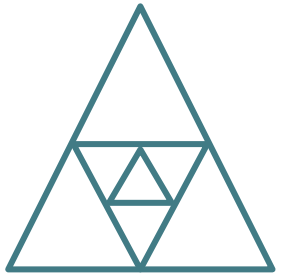
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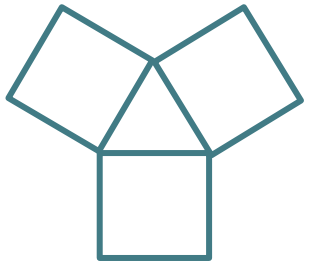
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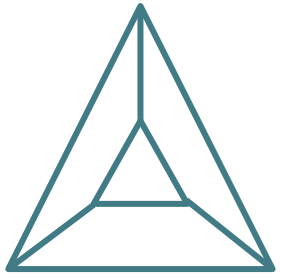
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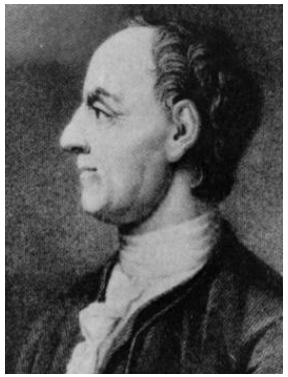


(7)



(8)

Leonhard Euler



The Father of Graph Theory
Leonhard Euler
(1707-1783)

How to pronounce “Leonhard Euler”?



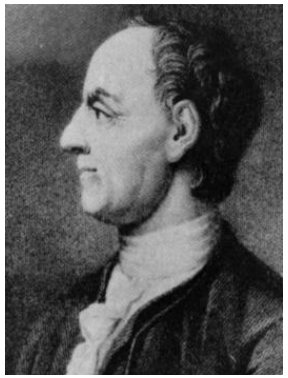
Swiss-German polymath

- Mathematician
- Physicist
- Astronomer
- Logician
- Geographer
- Music theorist
- Engineer

Identity: $e^{i\pi} + 1 = 0$
(The most beautiful equation in math.)

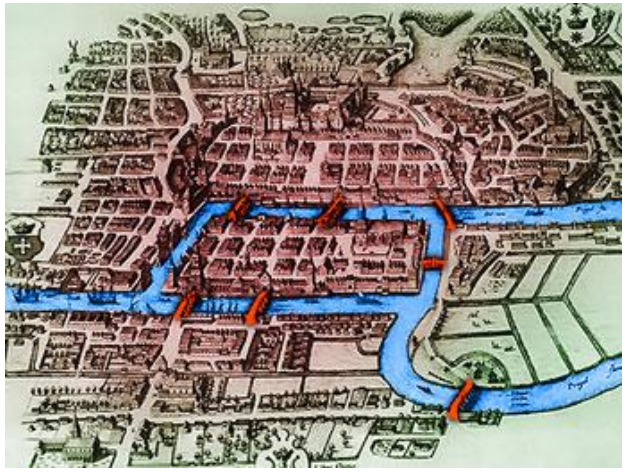
Königsberg Bridges

The subject of graph theory began in the year 1736 when the great mathematician Leonhard Euler published a paper giving the solution to the following puzzle:



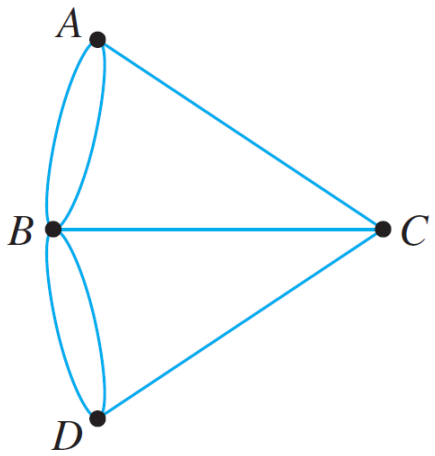
The Father of Graph Theory
Leonhard Euler
(1707-1783)

The town of Königsberg in Prussia (now Kaliningrad in Russia) was built at a point where two branches of the Pregel River came together. It consisted of an island and some land along the river banks. These were connected by 7 bridges.



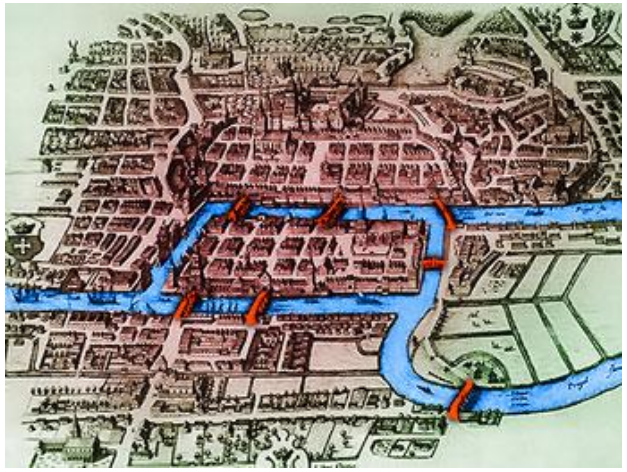
Euler asked: Is it possible to take a walk around town, starting and ending at the same location and crossing each of the 7 bridges exactly once?

Königsberg Bridges



In terms of this graph, the question is:

Is it possible to find a route through the graph that starts and ends at some vertex (*A*, *B*, *C*, or *D*) and traverses each edge exactly once?

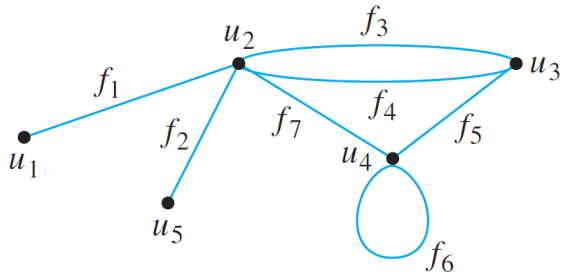


Euler asked: Is it possible to take a walk around town, starting and ending at the same location and crossing each of the 7 bridges exactly once?

Definitions

Travel in a graph is accomplished by moving from one vertex to another along a sequence of adjacent edges.

In the graph on the right, for instance, you can go from u_1 to u_4 by taking f_1 to u_2 and then f_7 to u_4. This is represented by writing $u_1 f_1 u_2 f_7 u_4$.



Walk, Trail, Path

Let G be a graph, and let v and w be vertices of G.

A **walk from v to w** is a finite alternating sequence of adjacent vertices and edges of G. Thus a walk has the form $v_0\, e_1\, v_1\, e_2 \ldots v_{n-1}\, e_n\, v_n$,

where the v's represent vertices, the e's represent edges, $v_0=v$, $v_n=w$, and for all $i \in \{1, 2, \ldots, n\}$, v_{i-1} and v_i are the endpoints of e_i. The number of edges, n, is the **length** of the walk.

The **trivial walk** from v to v consists of the single vertex v.

A **trail from v to w** is a walk from v to w that does not contain a repeated edge.

A **path from v to w** is a trail that does not contain a repeated vertex.

Definitions

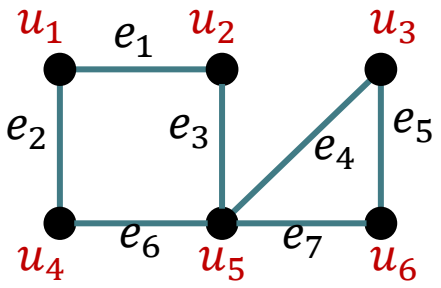
Closed Walk, Circuit/Cycle, Simple Circuit/Cycle, Cyclic

A **closed walk** is a walk that starts and ends at the same vertex.

A **circuit** (or **cycle**) is a closed walk of length at least 3 that does not contain a repeated edge.

A **simple circuit** (or **simple cycle**) is a circuit that does not have any other repeated vertex except the first and last.

An undirected graph is **cyclic** if it contains a loop or a cycle; otherwise, it is **acyclic**.



Examples:

$u_1e_1u_2e_3u_5e_4u_3e_5u_6e_7u_5e_3u_2$ is a walk (may repeat edges and/or vertices).

$u_1e_1u_2e_3u_5e_4u_3e_5u_6e_7u_5e_6u_4$ is a trail (must not repeat edges).

$u_1e_1u_2e_3u_5e_4u_3e_5u_6$ is a path (must not repeat vertices and edges).

$u_5e_6u_4e_2u_1e_1u_2e_3u_5e_7u_6e_5u_3e_4u_5$ is a circuit.

$u_5e_6u_4e_2u_1e_1u_2e_3u_5$ is a simple circuit.

Notes

Because most of the major developments in graph theory have happened relatively recently and in a variety of different contexts, the terms used in the subject have not been standardized.

Susanna Epp's book	Others
Graph	Multigraph
Simple graph	Graph
Vertex	Node
Edge	Arc
Trail	Path
Path	Simple path
Simple circuit	Cycle

The terminology in this book is among the most common, but if you consult other sources, be sure to check their definitions.

For CS1231S, we will follow the terminology in Epp's book.

Connectedness

A graph is connected if it is possible to travel from any vertex to any other vertex.

Definition: Connectedness

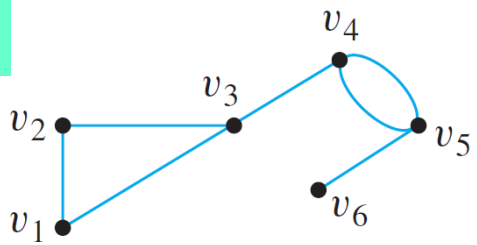
Two vertices *v* and *w* of a graph *G*=(*V*,*E*) are **connected** if and only if there is a walk from *v* to *w*.

The graph *G* is connected if and only if given *any* two vertices *v* and *w* in *G*, there is a walk from *v* to *w*. Symbolically,

G is connected iff $\forall$ vertices $v, w \in V$, $\exists$ a walk from *v* to *w*.

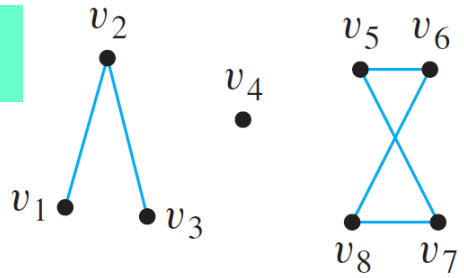
Example: Which of the following graphs are connected?

Yes

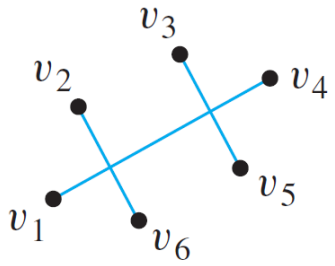


(a)

No



(b)



(c)

No

Connectedness

Some useful facts relating circuits and connectedness are collected in the following lemma.

Lemma 10.2.1

Let G be a graph.

a. If G is connected, then any two distinct vertices of G can be connected by a path.

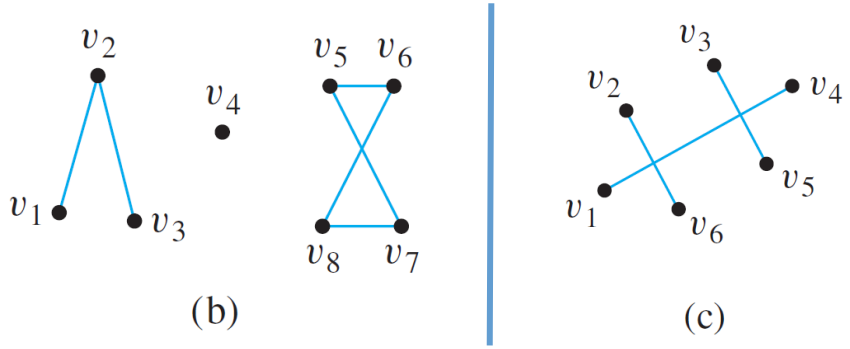
b. If vertices v and w are part of a circuit in G and one edge is removed from the circuit, then there still exists a trail from v to w in G.

c. If G is connected and G contains a circuit, then an edge of the circuit can be removed without disconnecting G.

Connected Component

The graphs in (*b*) and (*c*) are both made up of three pieces, each of which is itself a connected graph.

A *connected component* of a graph is a connected subgraph of <u>largest possible size</u>.



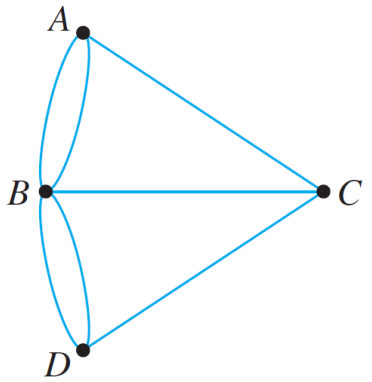
Definition: Connected Component

A graph *H* is a **connected component** of a graph *G* if and only if

1. The graph *H* is a subgraph of *G*;
2. The graph *H* is connected; and
3. No connected subgraph of *G* has *H* as a subgraph and contains vertices or edges that are not in *H*.

Euler Circuits

Now, let's go back to the puzzle of the Königsberg bridges.



Is it possible to find a route through the graph that starts and ends at some vertex and traverses each edge exactly once?

Definition: Euler Circuit

Let G be a graph. An **Euler circuit** for G is a circuit that contains every vertex and traverses every edge of G exactly once.

Definition: Eulerian Graph

An **Eulerian graph** is a graph that contains an Euler circuit.

Theorem 10.2.2

If a graph has an Euler circuit, then every vertex of the graph has positive even degree.

Contrapositive Version of Theorem 10.2.2

If some vertex of a graph has odd degree, then the graph does not have an Euler circuit.

Euler Circuits and Euler Trails

Theorem 10.2.3

If a graph G is connected and the degree of every vertex of G is a positive even integer, then G has an Euler circuit.

Theorem 10.2.4

A graph G has an Euler circuit if and only if G is connected and every vertex of G has positive even degree.

Definition: Euler Trail

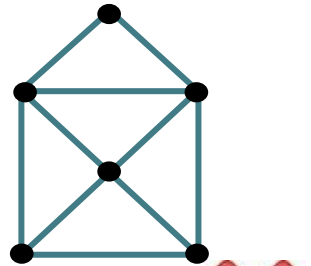
Let G be a graph, and let v and w be two distinct vertices of G. An **Euler trail/path from v to w** is a sequence of adjacent edges and vertices that starts at v, ends at w, passes through every vertex of G at least once, and traverses every edge of G exactly once.

Corollary 10.2.5

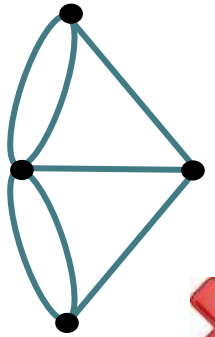
Let G be a graph, and let v and w be two distinct vertices of G. There is an Euler trail from v to w if and only if G is connected, v and w have odd degree, and all other vertices of G have positive even degree.

Euler Circuits and Euler Trails

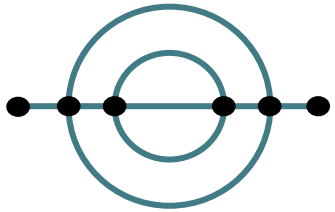
Does each of the following graphs have an Euler circuit?



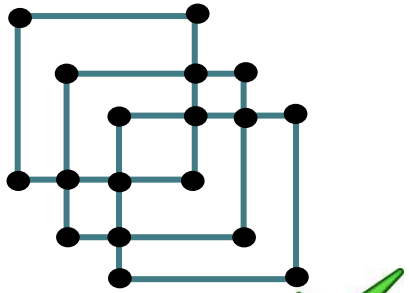
(1)



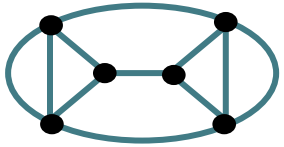
(2)



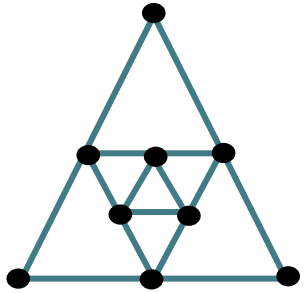
(3)



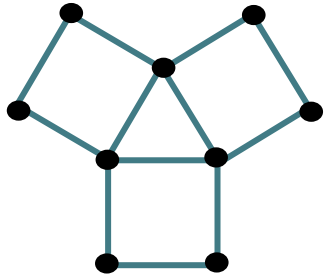
(4)



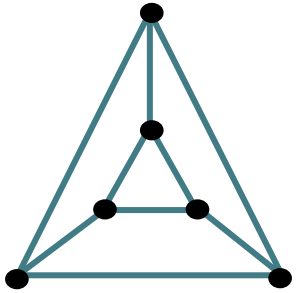
(5)



(6)



(7)

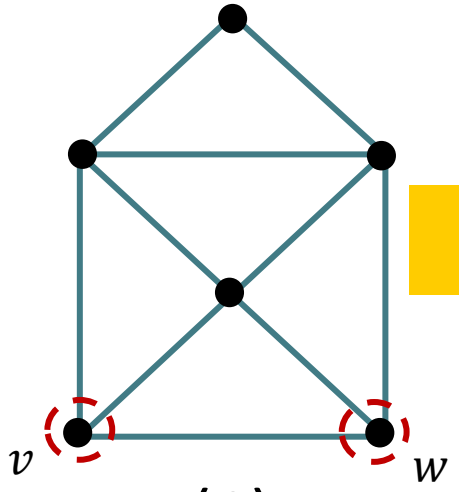


(8)

Which of them have an Euler trail?

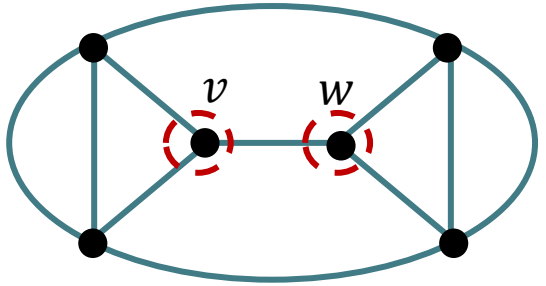
Euler Circuits and Euler Trails

The following graphs do not have an Euler circuit. Do they have an Euler trail?



Yes

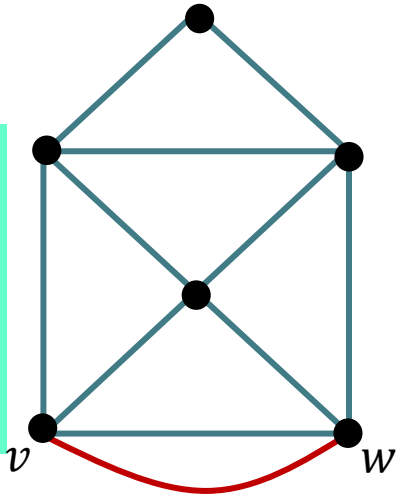
(1)

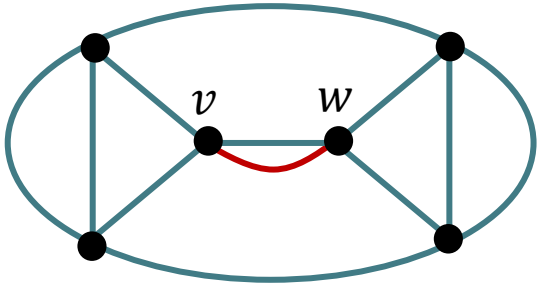


Yes

(5)

Adding an edge between the two vertices with odd degree will give us an Euler circuit.





Hamiltonian Circuits

A related question:
Given a graph *G*, is it possible to find a circuit for *G* in which all the *vertices* of *G* (except the first and the last) appear exactly once?

In 1859 the Irish mathematician Sir William Rowan Hamilton introduced a puzzle in the shape of a dodecahedron (DOH-dek-a-HEE-dron). (Figure 10.2.6 contains a drawing of a dodecahedron, which is a solid figure with 12 identical pentagonal faces.)

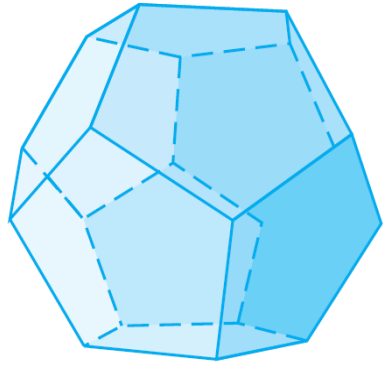


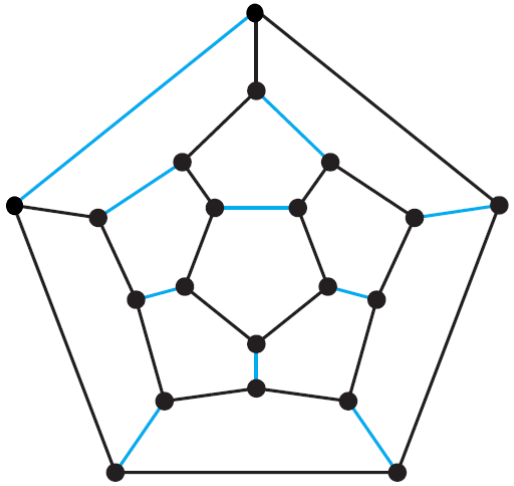
Figure 10.2.6 Dodecahedron

Each vertex was labeled with the name of a city — London, Paris, Singapore, New York, and so on.

The problem Hamilton posed was to start at one city and tour the world by visiting each other city exactly once and returning to the starting city.

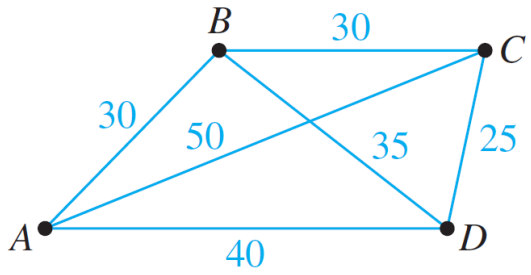
Hamiltonian Circuits

One way to solve the puzzle is to imagine the surface of the dodecahedron stretched out and laid flat in the plane, as follows:



The circuit denoted with black lines is one solution.
(Note that although every city is visited, many edges are omitted from the circuit.)

If we add values (called weights) to each edge, this becomes the travelling salesman problem.



Hamiltonian Circuits

Definition: Hamiltonian Circuit

Given a graph *G*, a **Hamiltonian circuit** for *G* is a simple circuit that includes every vertex of *G*. (That is, every vertex appears exactly once, except for the first and the last, which are the same.)

Definition: Hamiltonian Graph

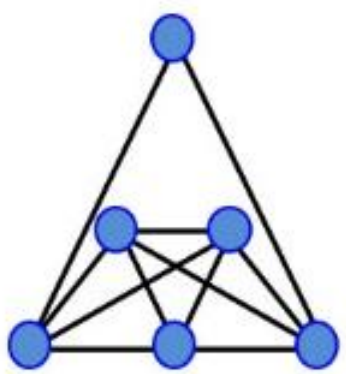
A **Hamiltonian graph** (also called **Hamilton graph**) is a graph that contains a Hamiltonian circuit.

Note that although an Euler circuit for a graph *G* must include every vertex of *G*, it may visit some vertices more than once and hence may not be a Hamiltonian circuit.

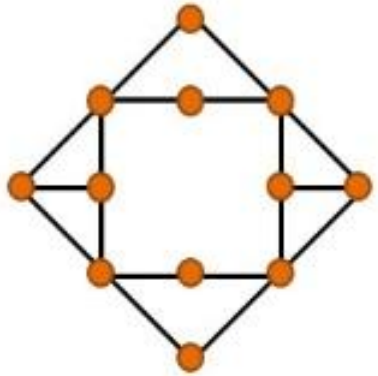
On the other hand, a Hamiltonian circuit for *G* does not need to include all the edges of *G* and hence may not be an Euler circuit.

Past-Year's Exam Question!

AY2019/20 Sem1



Graph *A*



Graph *B*

Which of the following statements is true?

A. Graphs *A* and *B* are both Eulerian and Hamiltonian.

B. Graph *A* is both Eulerian and Hamiltonian; graph *B* is neither Eulerian nor Hamiltonian.

C. Graph *A* is Eulerian but not Hamiltonian; graph *B* is neither Eulerian nor Hamiltonian.

D. Graph *A* is Eulerian but not Hamiltonian; graph *B* is Hamiltonian but not Eulerian.

E. Graphs *A* and *B* are Hamiltonian but not Eulerian.

Hamiltonian Circuits

Despite the analogous-sounding definitions of Euler and Hamiltonian circuits, the mathematics of the two are very different.

Theorem 10.2.4

A graph G has an Euler circuit if and only if G is connected and every vertex of G has positive even degree.

Determining whether a graph has an Euler circuit is easy – Theorem 10.2.4 gives a simple criterion.

Unfortunately, there is no analogous criterion for determining whether a given graph has a Hamiltonian circuit, nor is there even an efficient algorithm for finding such a circuit.

Hamiltonian Circuits

There is, however, a simple technique that can be used in many cases to show that a graph does *not* have a Hamiltonian circuit.

Proposition 10.2.6

If a graph G has a Hamiltonian circuit, then G has a subgraph H with the following properties:

1. H contains every vertex of G.
2. H is connected.
3. H has the same number of edges as vertices.
4. Every vertex of H has degree 2.

The contrapositive of Proposition 10.2.6 says that if a graph G does *not* have a subgraph H with properties (1)–(4), then G does *not* have a Hamiltonian circuit.

Adjacency Matrix of a Graph

Definition: Adjacency Matrix of a Directed Graph

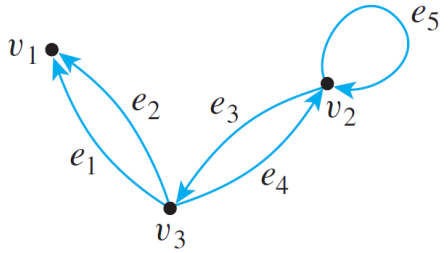
Let G be a directed graph with ordered vertices $v_1, v_2, \ldots v_n$. The **adjacency matrix of G** is the $n \times n$ matrix $\mathbf{A} = (a_{ij})$ over the set of non-negative integers such that

a_{ij} = the number of arrows from v_i to v_j for all $i, j = 1, 2, \ldots, n$.

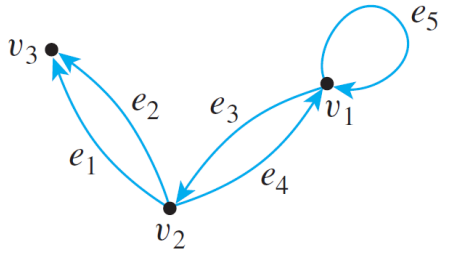
Example: Find the adjacency matrices of the two directed graphs below.

$$\begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \end{array} \begin{array}{c} \begin{array}{ccc} v_1 & v_2 & v_3 \end{array} \\ \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 2 & 1 & 0 \end{bmatrix} \end{array}$$

(a)



(a)



(b)

$$\begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \end{array} \begin{array}{c} \begin{array}{ccc} v_1 & v_2 & v_3 \end{array} \\ \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix} \end{array}$$

(b)

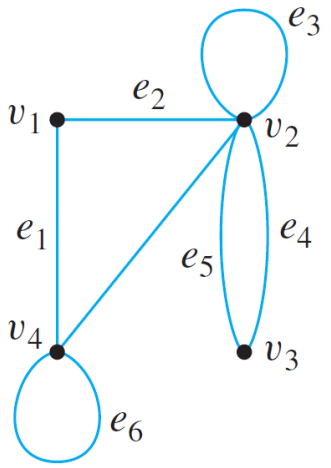
Adjacency Matrix of a Graph

Definition: Adjacency Matrix of an Undirected Graph

Let G be an undirected graph with ordered vertices $v_1, v_2, \ldots v_n$. The **adjacency matrix of G** is the $n \times n$ matrix $\mathbf{A} = (a_{ij})$ over the set of non-negative integers such that

a_{ij} = the number of edges connecting v_i and v_j for all $i, j = 1, 2, \ldots, n$.

Example: Find the adjacency matrix for the graph G shown below.



$$\mathbf{A} = \begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \end{array} \begin{array}{c} \begin{array}{cccc} v_1 & v_2 & v_3 & v_4 \end{array} \\ \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 1 & 2 & 1 \\ 0 & 2 & 0 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix} \end{array}$$

Note that the matrix is symmetric.

Definition: Symmetric Matrix

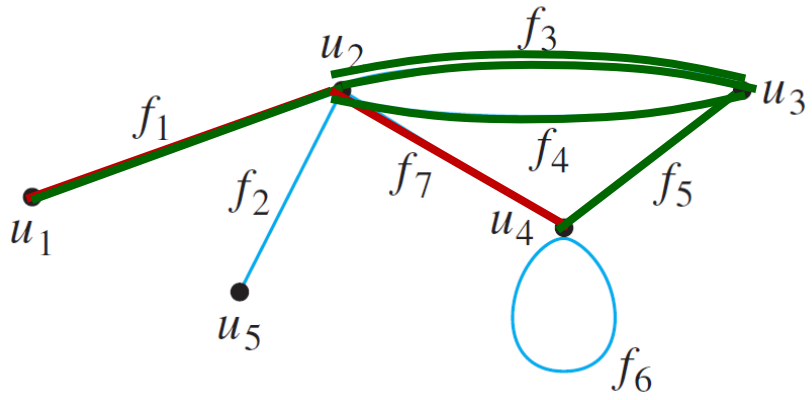
An $n \times n$ square matrix $A = (a_{ij})$ is called **symmetric** if, and only if, $a_{ij} = a_{ji}$ for all $i, j = 1, 2, \ldots, n$.

Counting Walks of Length N

A walk in a graph consists of an alternating sequence of vertices and edges.

The number of edges, possibly repeated, in the sequence is called the **length** of the walk.

For instance, the walk $v_2e_3v_3e_4v_2e_2v_2e_3v_3$ has length 4 (counting e_3 twice).

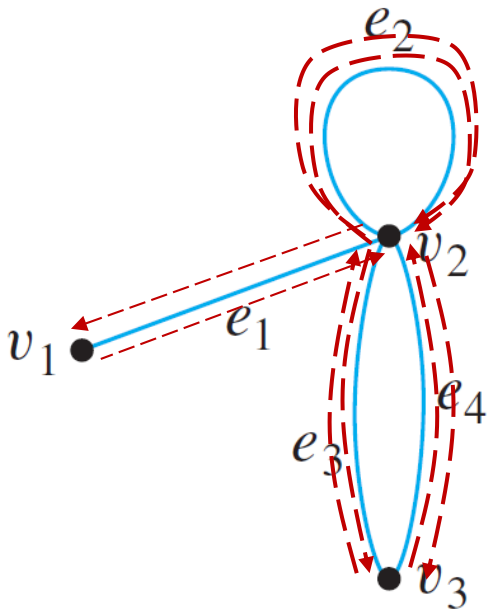


$u_1 f_1 u_2 f_7 u_4$ has length of 2.

$u_1 f_1 u_2 f_3 u_3 f_4 u_2 f_3 u_3 f_5 u_4$ has length of 5.

Counting Walks of Length N

Example: Consider the following graph G.
How many distinct walks of length 2 connect v_2 and v_2?



One walk of length 2 from v_2 to v_2 via v_1:
$v_2e_1v_1e_1v_2$.

One walk of length 2 from v_2 to v_2 via v_2:
$v_2e_2v_2e_2v_2$.

Four walks of length 2 from v_2 to v_2 via v_3:
$v_2e_3v_3e_4v_2$,
$v_2e_4v_3e_3v_2$,
$v_2e_3v_3e_3v_2$,
$v_2e_4v_3e_4v_2$.

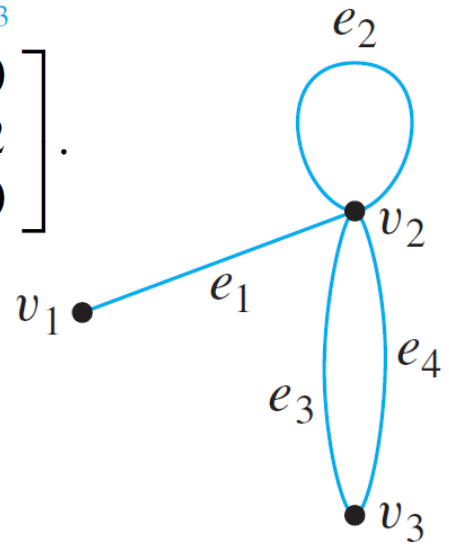
Total = 6

Counting Walks of Length N

The general question of finding the number of walks that have a given length and connect two particular vertices of a graph can easily be answered using **matrix multiplication**.

Consider the adjacency matrix **A** of the graph G.

$$\mathbf{A} = \begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \end{array} \begin{array}{c} \begin{array}{ccc} v_1 & v_2 & v_3 \end{array} \\ \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 2 \\ 0 & 2 & 0 \end{bmatrix} \end{array}.$$



Compute $\mathbf{A}^2$:

$$\begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 2 \\ 0 & 2 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 2 \\ 0 & 2 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 2 \\ 1 & 6 & 2 \\ 2 & 2 & 4 \end{bmatrix}.$$

Note that the entry in row 2 and column 2 is **6**, which equals the number of walks of length 2 from v_2 to v_2.

Reason: To compute a_{22}, you multiply row 2 of **A** with column 2 of **A** to obtain a sum of three terms:

$$\begin{bmatrix} 1 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = 1 \cdot 1 + 1 \cdot 1 + 2 \cdot 2.$$

Counting Walks of Length N

Theorem 10.3.2

If G is a graph with vertices $v_1, v_2, \ldots, v_m$ and $\mathbf{A}$ is the adjacency matrix of G, then for each positive integer n and for all integers $i, j = 1, 2, \ldots, m$,
the ij-th entry of $\mathbf{A}^n$ = the number of walks of length n from v_i to v_j.

Isomorphisms of Graphs

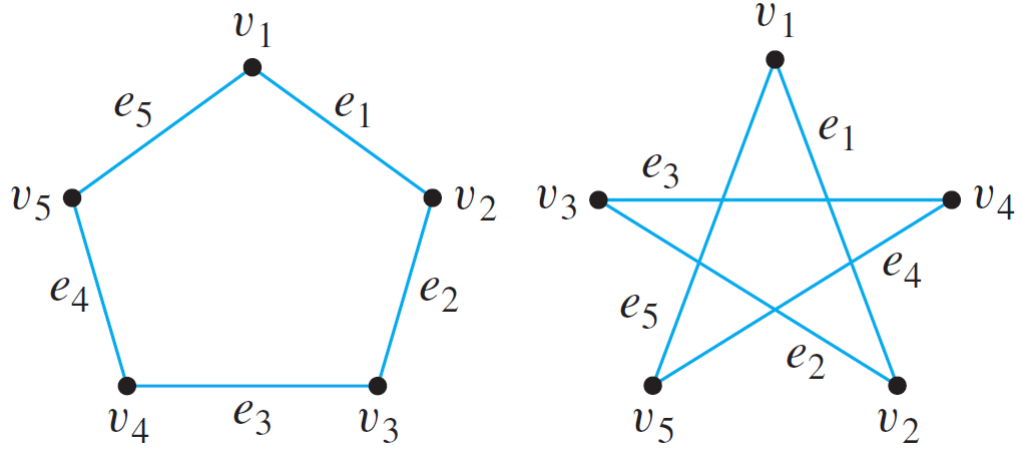


Figure 10.4.1

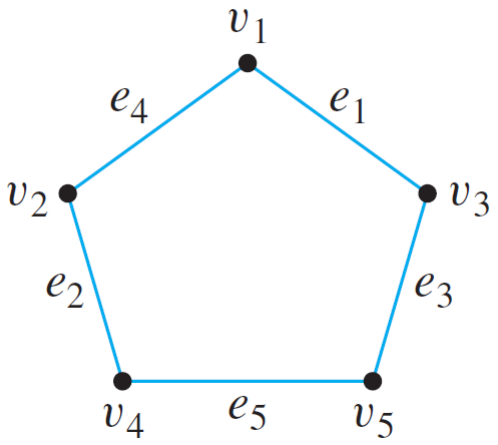
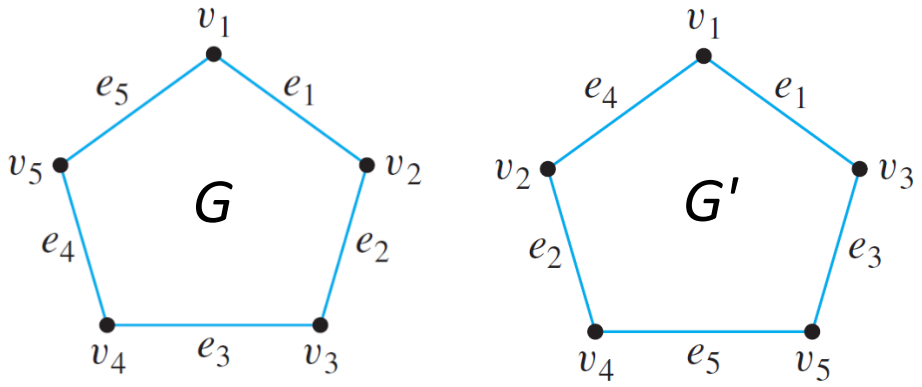


Figure 10.4.2

These 3 graphs are isomorphic to one another.

Isomorphisms of Graphs



If the vertices and edges of G are relabeled by the functions shown below, then G' becomes the same as G.

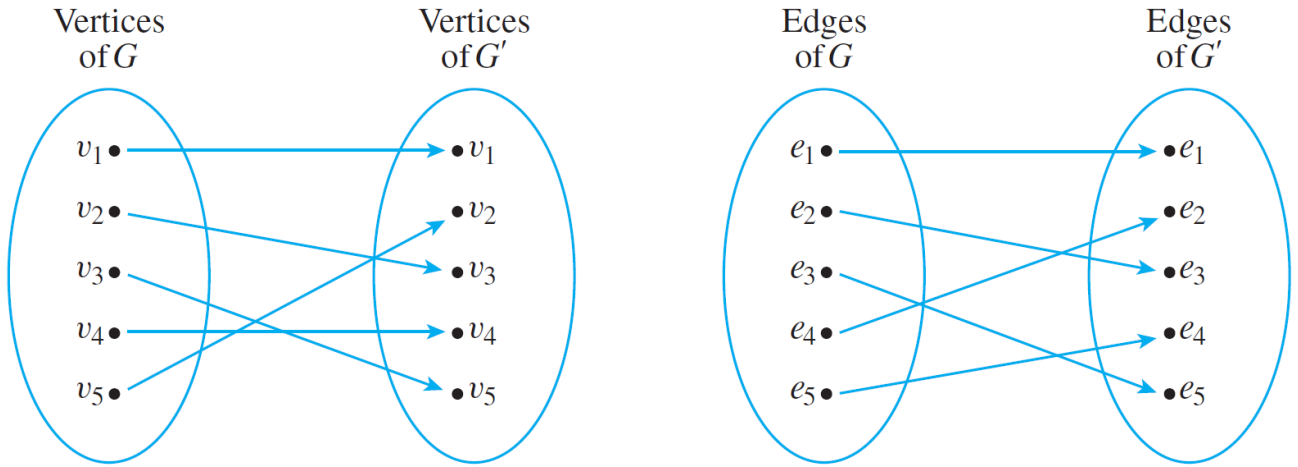


Figure 10.4.3

Note that these relabeling functions are **bijective**.

Isomorphisms of Graphs

Two graphs G and G' that are the same except for the labeling of their vertices and edges are called *isomorphic*. In other words, there exists matching between the vertices such that two vertices are connected by an edge in G if and only if corresponding vertices are connected by an edge in G'.

Definition: Isomorphic Graph

Let $G = (V_G, E_G)$ and $G' = (V_{G'}, E_{G'})$ be two graphs.

G is isomorphic to G', denoted $G \cong G'$, if and only if there exist bijections $g: V_G \to V_{G'}$ and $h: E_G \to E_{G'}$ that preserve the edge-endpoint functions of G and G' in the sense that for all $v \in V_G$ and $e \in E_G$,

v is an endpoint of $e \Leftrightarrow g(v)$ is an endpoint of $h(e)$.

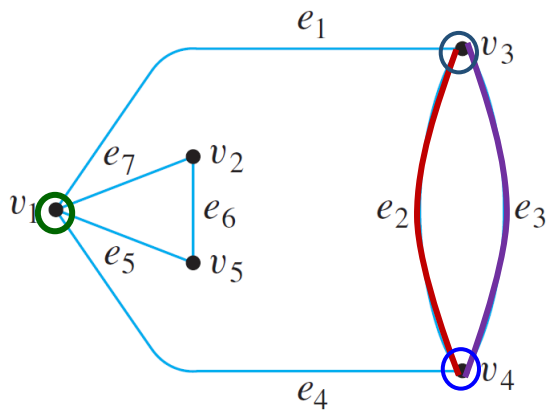
Alternative definition (for simple graphs)

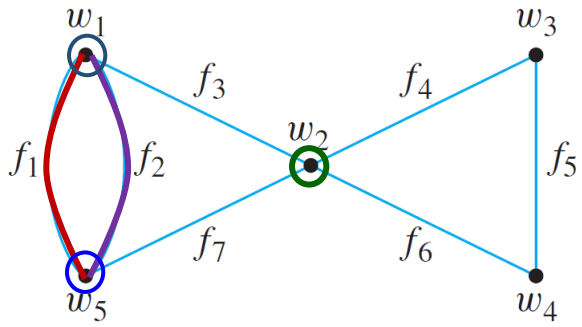
Let $G = (V_G, E_G)$ and $G' = (V_{G'}, E_{G'})$ be two simple graphs.

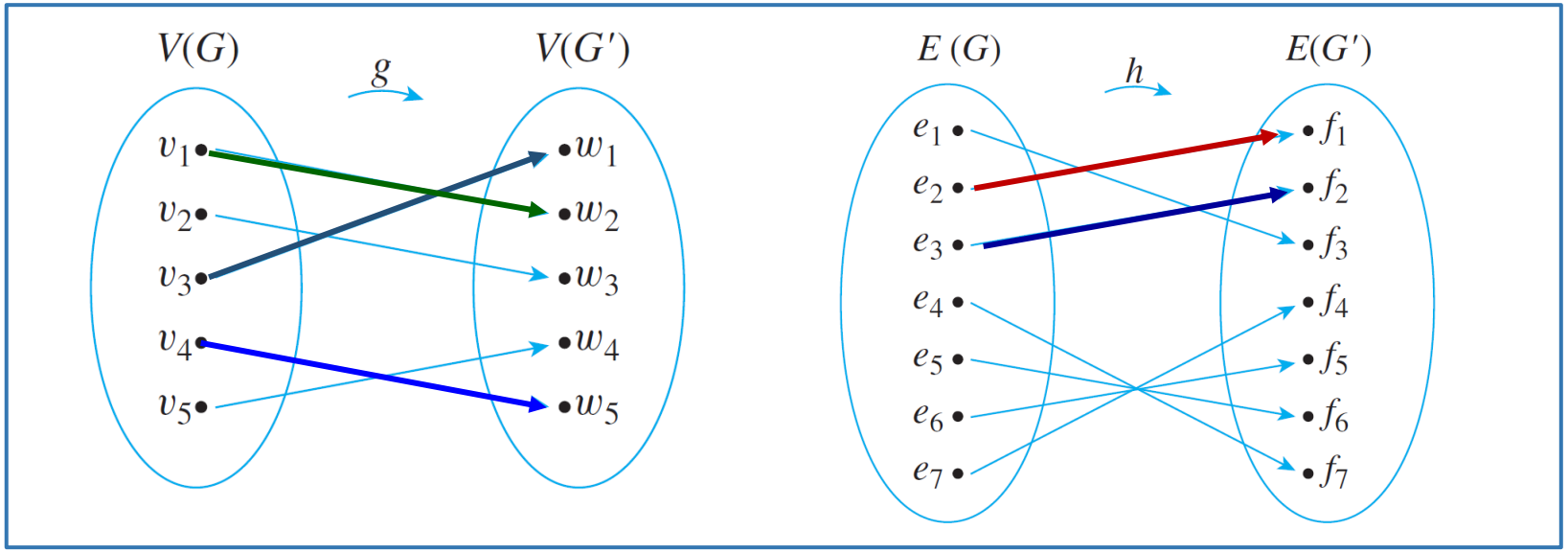
G is isomorphic to G' if and only if there exists a permutation $\pi: V_G \to V_{G'}$ such that $\{u, v\} \in E_G \Leftrightarrow \{\pi(u), \pi(v)\} \in E_{G'}$.

Isomorphisms of Graphs

Example: Show that these two graphs are isomorphic.







Isomorphisms of Graphs

It is not hard to show that graph isomorphism is an equivalence relation on a set of graphs; in other words, it is reflexive, symmetric, and transitive.

Theorem 10.4.1 Graph Isomorphism is an Equivalence Relation

Let S be a set of graphs and let $\cong$ be the relation of graph isomorphism on S. Then $\cong$ is an equivalence relation on S.

Exercise: Prove that graph isomorphism $\cong$ is an equivalence relation.

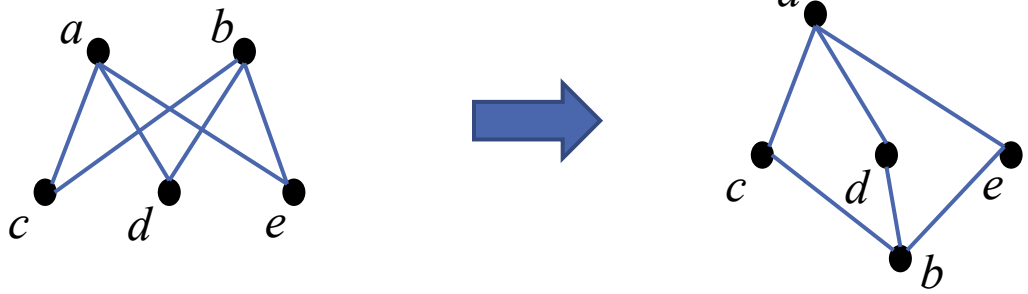
Planar Graph

Definition: Planar Graph

A **planar graph** is a graph that can be drawn on a (two-dimensional) plane without edges crossing.

Is Figure 10.4.4 a planar graph?

Yes, it is a planar graph.

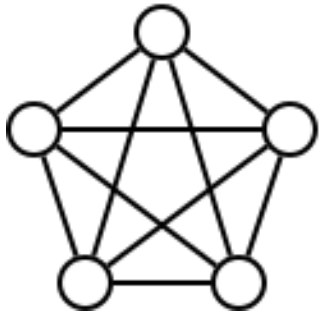


Non-planar representation of the graph

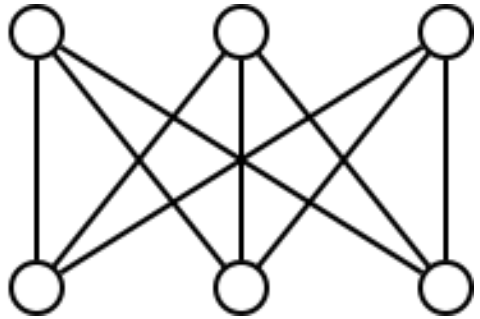
Planar representation of the graph

Planar Graph

The smallest non-planar graphs



K_5



$K_{3,3}$



Kazimierz Kuratowski
(1896 – 1980)
Polish mathematician
and logician

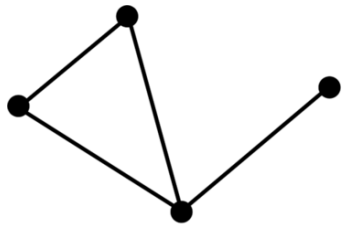
Kuratowski's Theorem:

A finite graph is planar if and only if it does not contain a subgraph that is a subdivision of K_5 or $K_{3,3}$.

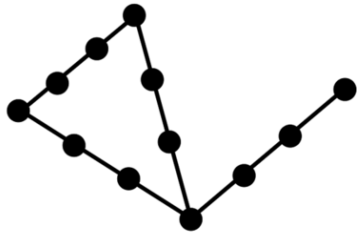
Subdivision of a Graph

Given a graph G, a subdivision of G, is a new graph obtained by inserting new vertices into the edges of G.

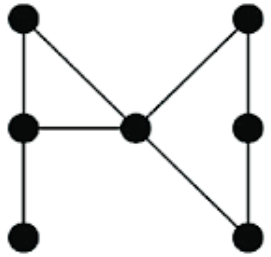
Typically done by removing an edge, adding a new vertex, and connecting the new vertex to the original edge's two endpoints.



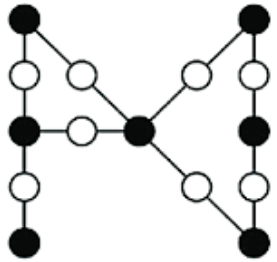
G_1



G_1': subdivision of G_1



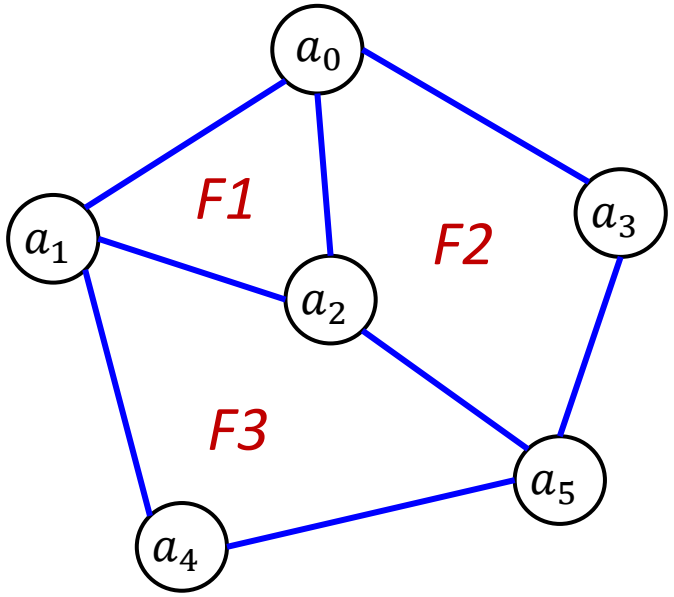
G_2



G_2': subdivision of G_2

Euler's Formula

When we draw a planar representation of a planar graph, it divides the plane up into regions or faces.



(*F4* = 'outside' the planar graph)

Euler's Formula

For a connected planar simple graph $G = (V, E)$ with $e = |E|$ and $v = |V|$, if we let f be the number of faces, then

$$f = e - v + 2$$

$e = 8$

$v = 6$

$f = 8 - 6 + 2 = 4$

Past-Year's Exam Question!

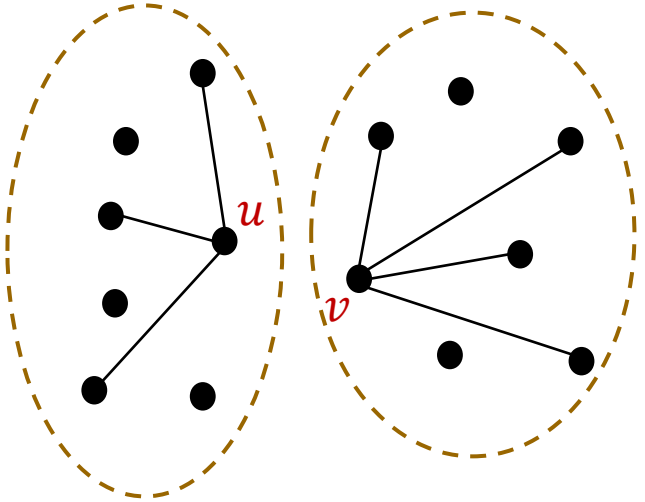
AY2016/17 Sem1

Let G be a simple graph with n vertices where every vertex has degree at least $\left\lfloor \frac{n}{2} \right\rfloor$. Prove that G is connected.

Proof by contradiction:

1. Suppose G is not connected.
2. Let u and v be the vertices in two separate connected components.
3. Then the number of vertices in the union of their neighborhood, including u and v, is at least $\left\lfloor \frac{n}{2} \right\rfloor + \left\lfloor \frac{n}{2} \right\rfloor + 2 \geq n + 1$.
4. But this exceeds the number of vertices in the graph. Hence, G must be connected.

G is not connected



Past-Year's Exam Question!

AY2022/23 Sem2 Q22

Cycle Graph:

A cycle graph C_n , $n \geq 3$, is an undirected graph consisting of n vertices $v_1, v_2, \ldots, v_n$ and n edges $\{v_1, v_2\}, \{v_2, v_3\}, \ldots, \{v_{n-1}, v_n\}, \{v_n, v_1\}$.

Star Graph:

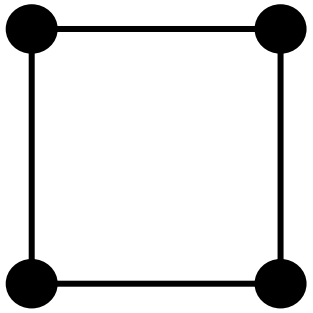
A star graph S_n , $n \geq 3$, is an undirected graph consisting of $n + 1$ vertices $v_1, v_2, \ldots, v_n, v_{n+1}$ and n edges $\{v_1, v_{n+1}\}, \{v_2, v_{n+1}\}, \ldots, \{v_{n-1}, v_{n+1}\}, \{v_n, v_{n+1}\}$.

Wheel Graph:

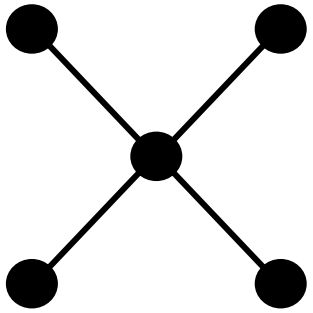
A wheel graph W_n , $n \geq 3$, is defined as follows: the vertex set of W_n is the vertex set of S_n and the edge set of W_n is the union of the edge set of C_n and the edge set of S_n.

(a) Which of the following are complete graphs: C_4, S_4, W_4?

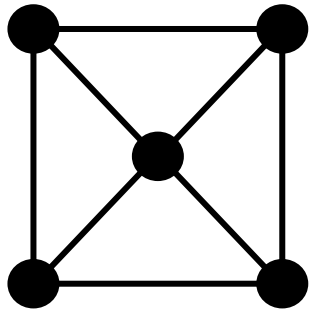
C_4



S_4



W_4



None.

Past-Year's Exam Question!

AY2022/23 Sem2 Q22

Cycle Graph:

A cycle graph C_n , $n \geq 3$, is an undirected graph consisting of n vertices $v_1, v_2, \ldots, v_n$ and n edges $\{v_1, v_2\}, \{v_2, v_3\}, \ldots, \{v_{n-1}, v_n\}, \{v_n, v_1\}$.

Star Graph:

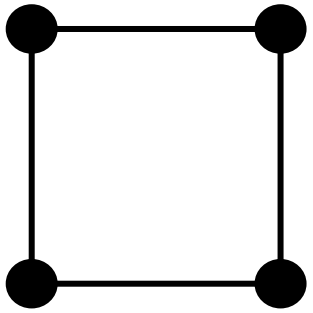
A star graph S_n , $n \geq 3$, is an undirected graph consisting of $n + 1$ vertices $v_1, v_2, \ldots, v_n, v_{n+1}$ and n edges $\{v_1, v_{n+1}\}, \{v_2, v_{n+1}\}, \ldots, \{v_{n-1}, v_{n+1}\}, \{v_n, v_{n+1}\}$.

Wheel Graph:

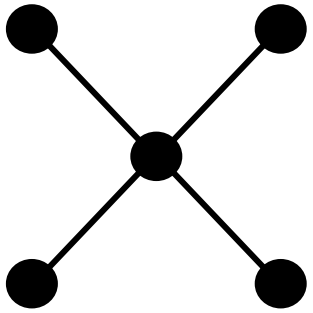
A wheel graph W_n , $n \geq 3$, is defined as follows: the vertex set of W_n is the vertex set of S_n and the edge set of W_n is the union of the edge set of C_n and the edge set of S_n.

(b) Which of the following are planar graphs: $C_{1231}, S_{1231}, W_{1231}$?

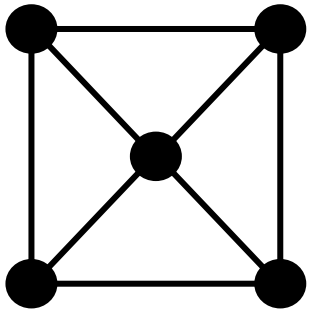
C_4



S_4



W_4



All.

C_4, S_4, W_4 shown for illustration.
Generalise to bigger graphs.

Past-Year's Exam Question!

AY2022/23 Sem2 Q22

Cycle Graph:

A cycle graph C_n , $n \geq 3$, is an undirected graph consisting of n vertices $v_1, v_2, \ldots, v_n$ and n edges $\{v_1, v_2\}, \{v_2, v_3\}, \ldots, \{v_{n-1}, v_n\}, \{v_n, v_1\}$.

Star Graph:

A star graph S_n , $n \geq 3$, is an undirected graph consisting of $n + 1$ vertices $v_1, v_2, \ldots, v_n, v_{n+1}$ and n edges $\{v_1, v_{n+1}\}, \{v_2, v_{n+1}\}, \ldots, \{v_{n-1}, v_{n+1}\}, \{v_n, v_{n+1}\}$.

Wheel Graph:

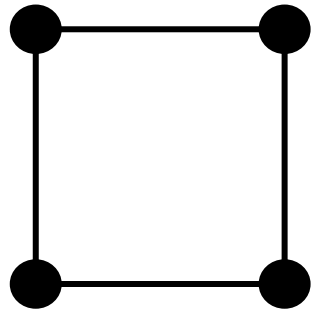
A ... ows: ... e ... of C_n

C_{1231} has an Euler circuit.

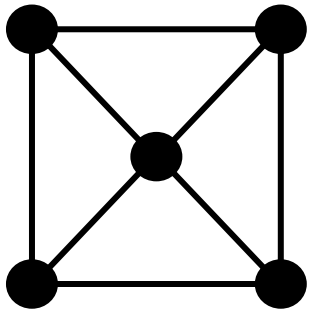
W_{1231} does not have an Euler circuit.

(c) Does C_{1231} have an Euler circuit? Does W_{1231} have an Euler circuit?

C_4



W_4



Degree of each vertex in C_n is 2; degree of each vertex along the boundary of W_n is 3.

Past-Year's Exam Question!

AY2022/23 Sem2 Q22

Cycle Graph:

A cycle graph C_n , $n \geq 3$, is an undirected graph consisting of n vertices $v_1, v_2, \ldots, v_n$ and n edges $\{v_1, v_2\}, \{v_2, v_3\}, \ldots, \{v_{n-1}, v_n\}, \{v_n, v_1\}$.

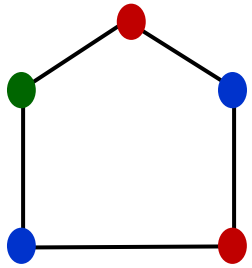
Star Graph:

A star graph S_n , $n \geq 3$, is an undirected graph consisting of $n + 1$ vertices $v_1, v_2, \ldots, v_n, v_{n+1}$ and n edges $\{v_1, v_{n+1}\}, \{v_2, v_{n+1}\}, \ldots, \{v_{n-1}, v_{n+1}\}, \{v_n, v_{n+1}\}$.

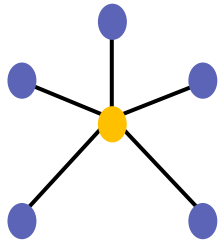
Wheel Graph:

A wheel graph W_n , $n \geq 3$, is defined as follows: the vertex set of W_n is the vertex set of S_n and the edge set of W_n is the union of the edge set of C_n and the edge set of S_n.

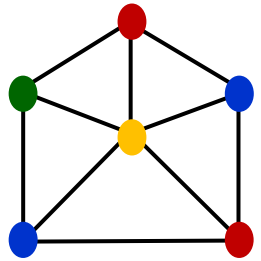
(d) Determine the minimum number of colours needed to colour C_5, S_5 and W_5 such that no two adjacent vertices have the same colour.



3 colours.



2 colours.

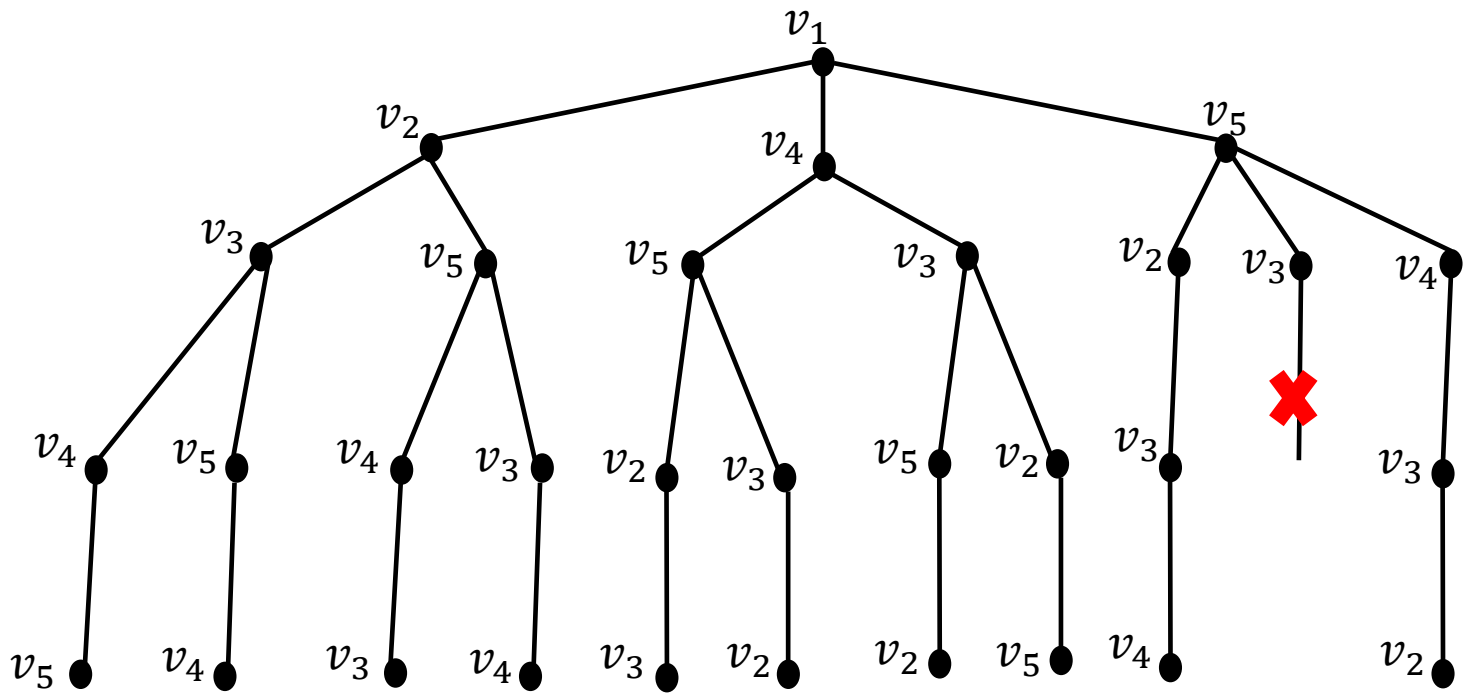


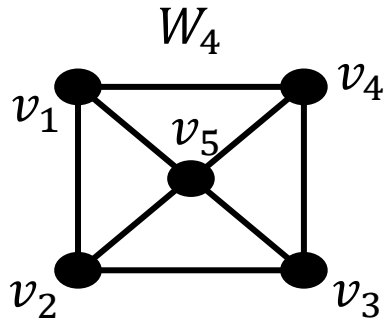
4 colours.

Past-Year's Exam Question!

AY2022/23 Sem2 Q22

(f) **Hamiltonian circuit** is a closed-loop, i.e., it starts and ends at the same vertex while visiting every other vertex in the graph at most once. On the other hand, a **Hamiltonian path** is an open-loop and it visits each vertex only once. In summary, Hamiltonian path is an open-loop Hamiltonian circuit with different start and end vertices and must visit each vertex only once. Draw the possibility tree to determine the number of Hamiltonian paths in W_4, assuming that the initial vertex is fixed as v_1.





(g) Find the number of unique open-loop Hamiltonian paths in W_4.

10 open-loop Hamiltonian paths.

Past-Year's Exam Question!

AY2022/23 Sem2 Q22

(j) Consider an undirected graph G whose connected components are $H_1, H_2, H_3, \cdots, H_k$ where $k > 1$.

Suppose $G = (V, E)$ and $H_i = (V_i, E_i)$ for $1 \leq i \leq k$.

Is $\{V_1, V_2, \cdots, V_k\}$ always a partition of V?.

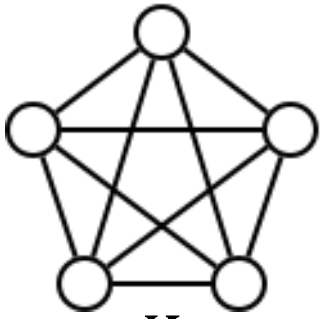
$\{V_1, V_2, \cdots, V_k\}$ is always a partition of V.

Is $\{E_1, E_2, \cdots, E_k\}$ always a partition of E?

$\{E_1, E_2, \cdots, E_k\}$ may not be a partition of E as some E_i could be $\emptyset$.

Past-Year's Exam Question!

AY2022/23 Sem1



18. Which of the following statements are true?

(i) $K_5 - \{e\}$ is planar where e is one of the edges in K_5.

(ii) K_5 is both Eulerian and Hamiltonian.

(iii) There is a simple planar graph with 6 vertices, 8 edges, and 7 faces.

A. Only (i).

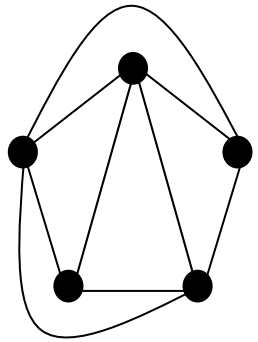
B. Only (i) and (ii).

C. Only (i) and (iii).

D. All of (i), (ii) and (iii).

E. None of the above.

(i)



(iii) By Euler's formula, $f = e - v + 2$ and $7 \neq 8 - 6 + 2$.

Past-Year's Exam Question!

AY2022/23 Sem1

19. Which of the following statements are true?

(i) Let G be a simple, undirected graph with 6 vertices, 5 edges and no cycles. Then it is not possible to have only one vertex in G with degree 1.

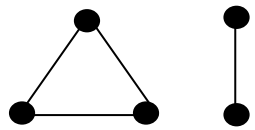
(ii) Let G be a simple, undirected graph with n vertices and e edges. If G is not connected, then $e < (n-1)$.

(iii) All cycles in the graph $K_{3,4}$ have an even number of edges.

A. Only (i).
B. Only (i) and (ii).
C. Only (i) and (iii).
D. All of (i), (ii) and (iii).
E. None of the above.

(i) G is a tree with 6 vertices. A tree with 2 or more vertices has at least 2 vertices with degree 1.

(ii) See figure below. G is not connected but $n = 5$, $e = 4$, so $e \not< (n-1)$.



(iii) $K_{3,4}$ is a complete bipartite graph. To complete a cycle one must traverse to the other set then back.

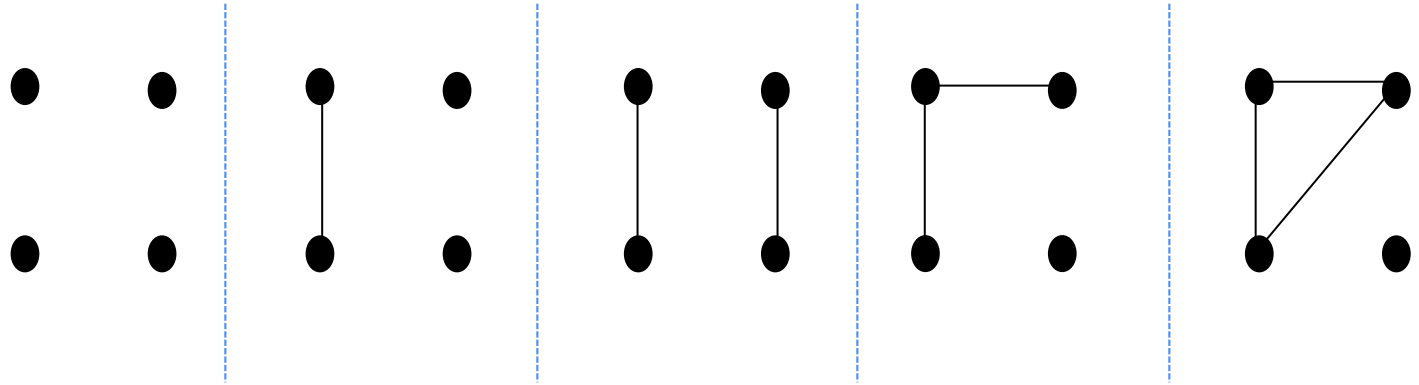
Past-Year's Exam Question!

AY2022/23 Sem1

20. How many non-isomorphic simple, undirected, disconnected graphs with 4 vertices are there?

A. 2.
B. 3.
C. 4.
D. 5.
E. 6.

5 graphs (0 edge: 1; 1 edge: 1; 2 edges: 2;
3 edges: 1; ≥4 edges: 0.)



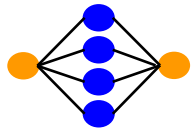
Past-Year's Exam Question!

AY2021/22 Sem2

Kuratowski's Theorem: A finite graph is planar if and only if it does not contain a subgraph that is a subdivision of K_5 or $K_{3,3}$.

20(b) Prove or disprove the following statements:

(i) The graph $K_{2,4}$ is a planar graph. True



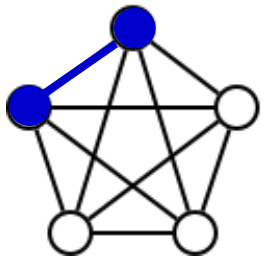
(ii) The graph $K_{3,4}$ is a planar graph. False. $K_{3,4}$ contains $K_{3,3}$.

(iii) There exists a simple graph that is connected and planar with 5 vertices and 8 faces.

False. Using Euler's formula ($f = e - v + 2$), the graph has $e = f + v - 2 = 11$ edges. But a simple graph on 5 vertices has at most $\binom{5}{2} = 10$ edges.

(iv) The graph $K_5 - \{e\}$ where e is any edge in the graph is 4-colourable.

True: Let $e = \{x, y\}$. Colour both x and y with the same colour, then the other vertices can be coloured with 3 other colours.



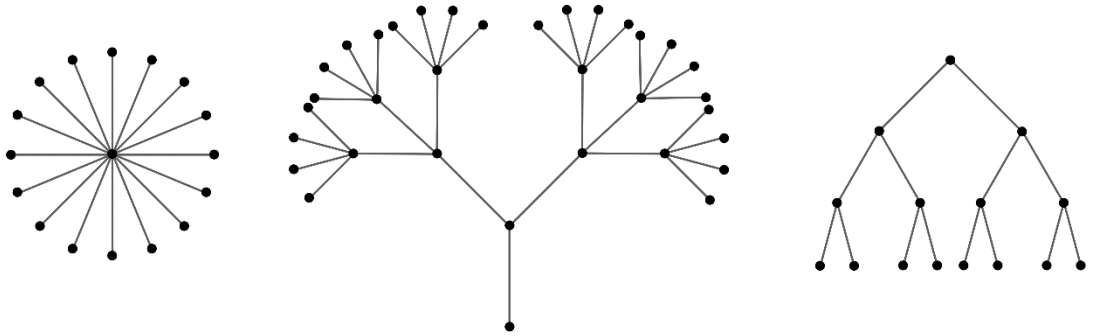
Reminder

Assignment #2 deadline:

Monday, 2 November 2026, 1pm

Next week's lectures

Trees



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